In [ ]:
```
from typing import Any, Dict, List, Tuple
import argparse
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
from sklearn.model_selection import GridSearchCV, StratifiedKFold
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
```

In [ ]:
```
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
df.head()
```

Out[ ]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns

In [ ]:
```
df.shape
```

Out[ ]:
```
(7043, 21)
```

In [ ]:
```
df.columns
```

Out[ ]:
```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   customerID        7043 non-null   object 
 1   gender            7043 non-null   object 
 2   SeniorCitizen     7043 non-null   int64  
 3   Partner           7043 non-null   object 
 4   Dependents        7043 non-null   object 
 5   tenure            7043 non-null   int64  
 6   PhoneService      7043 non-null   object 
 7   MultipleLines     7043 non-null   object 
 8   InternetService   7043 non-null   object 
 9   OnlineSecurity    7043 non-null   object 
 10  OnlineBackup      7043 non-null   object 
 11  DeviceProtection  7043 non-null   object 
 12  TechSupport       7043 non-null   object 
 13  StreamingTV       7043 non-null   object 
 14  StreamingMovies   7043 non-null   object 
 15  Contract          7043 non-null   object 
 16  PaperlessBilling  7043 non-null   object 
 17  PaymentMethod     7043 non-null   object 
 18  MonthlyCharges    7043 non-null   float64 
 19  TotalCharges      7043 non-null   object 
 20  Churn             7043 non-null   object 
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

In [ ]:
```
df.isnull().sum()
```

Out[ ]:
```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

In [ ]:
```
import pandas as pd
import numpy as np

from sklearn.model_selection import (
    RandomizedSearchCV,
    StratifiedKFold,
```

```
    train_test_split,
)
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.metrics import accuracy_score, confusion_matrix, classification_rep
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC

from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline as ImbPipeline

# Try to import XGBoost; if not installed, we will skip it
try:
    from xgboost import XGBClassifier
    xgb_available = True
except ImportError:
    XGBClassifier = None
    xgb_available = False

import matplotlib.pyplot as plt
import seaborn as sns

# Configure plots
sns.set(style="whitegrid")

# Display all columns
pd.set_option("display.max_columns", None)

print("Libraries imported. XGBoost available:", xgb_available)
```

```
Libraries imported. XGBoost available: True
```

In [ ]:

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.metrics import accuracy_score, confusion_matrix, classification_rep
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC

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# Try to import XGBoost; if not installed, we will skip it
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except ImportError:
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    xgb_available = False

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import seaborn as sns

sns.set(style="whitegrid")
pd.set_option("display.max_columns", None)
```

```
print("Libraries imported. XGBoost available:", xgb_available)
```

```
Libraries imported. XGBoost available: True
```

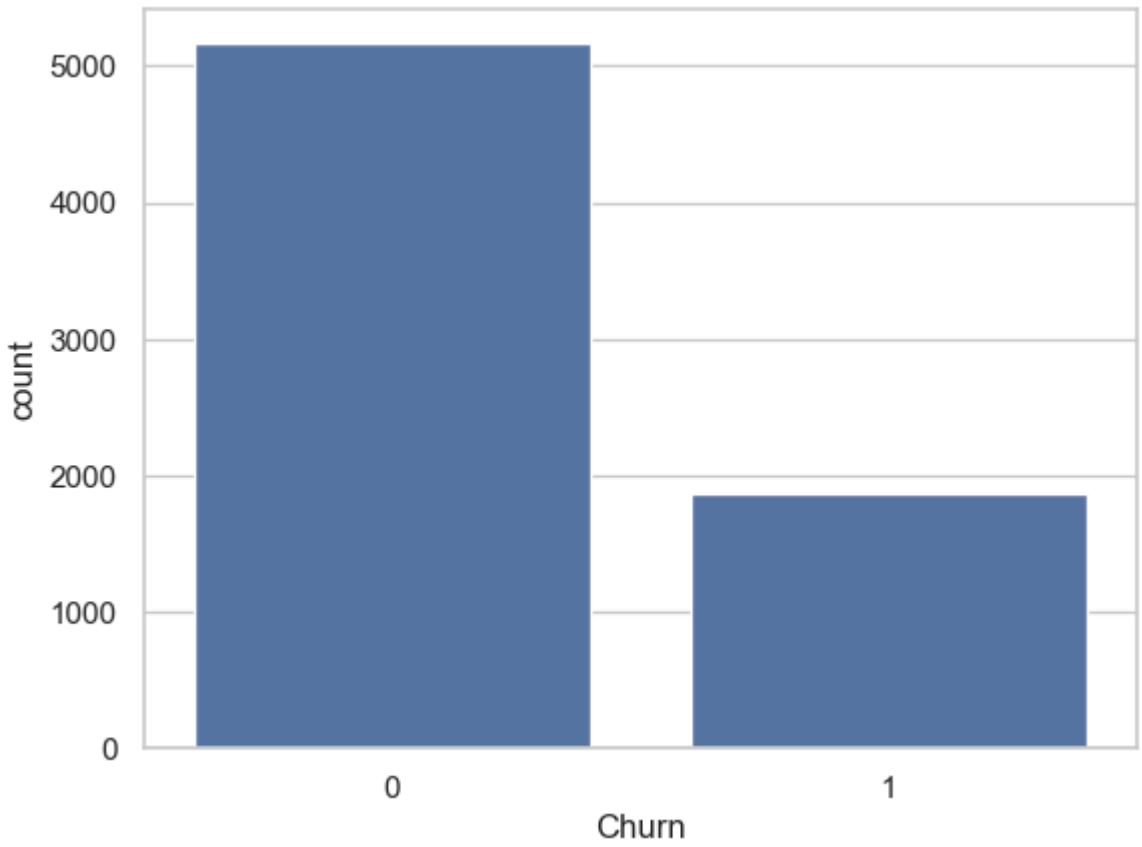
In [ ]:
```
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
df.head()
```

Out[ ]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mul
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

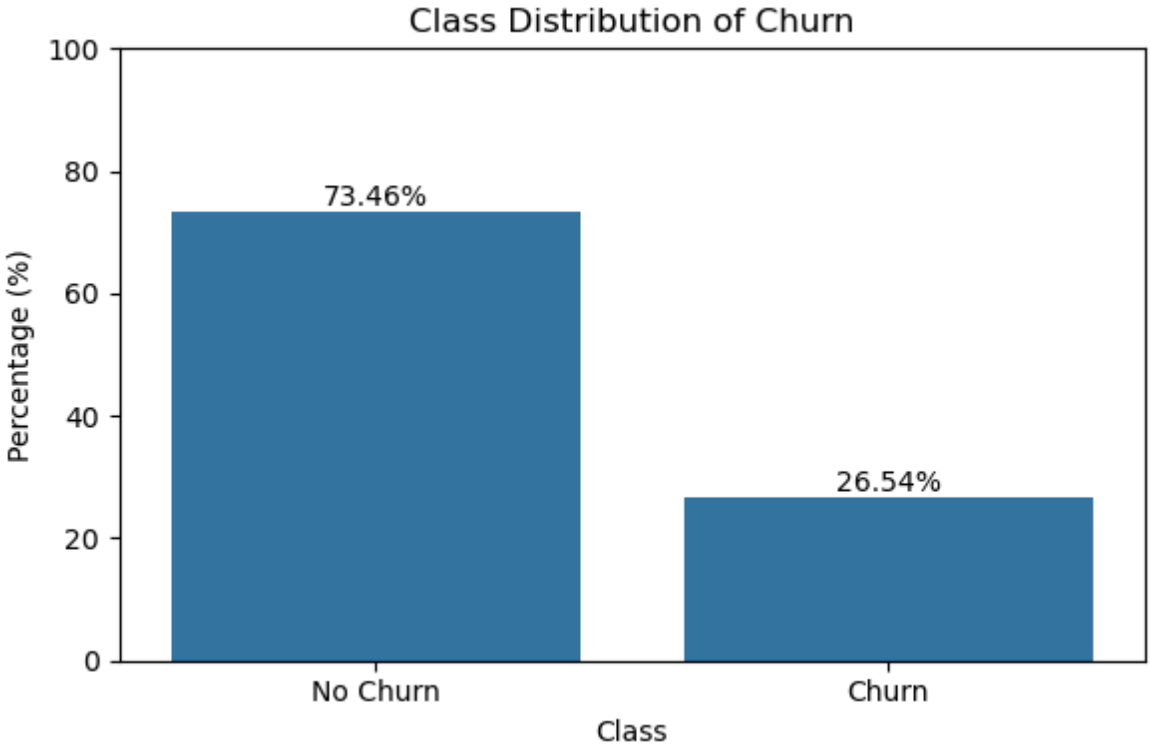
In [ ]:
```
sns.countplot(x='Churn', data=df)
```

Out[ ]:
```
<Axes: xlabel='Churn', ylabel='count'>
```



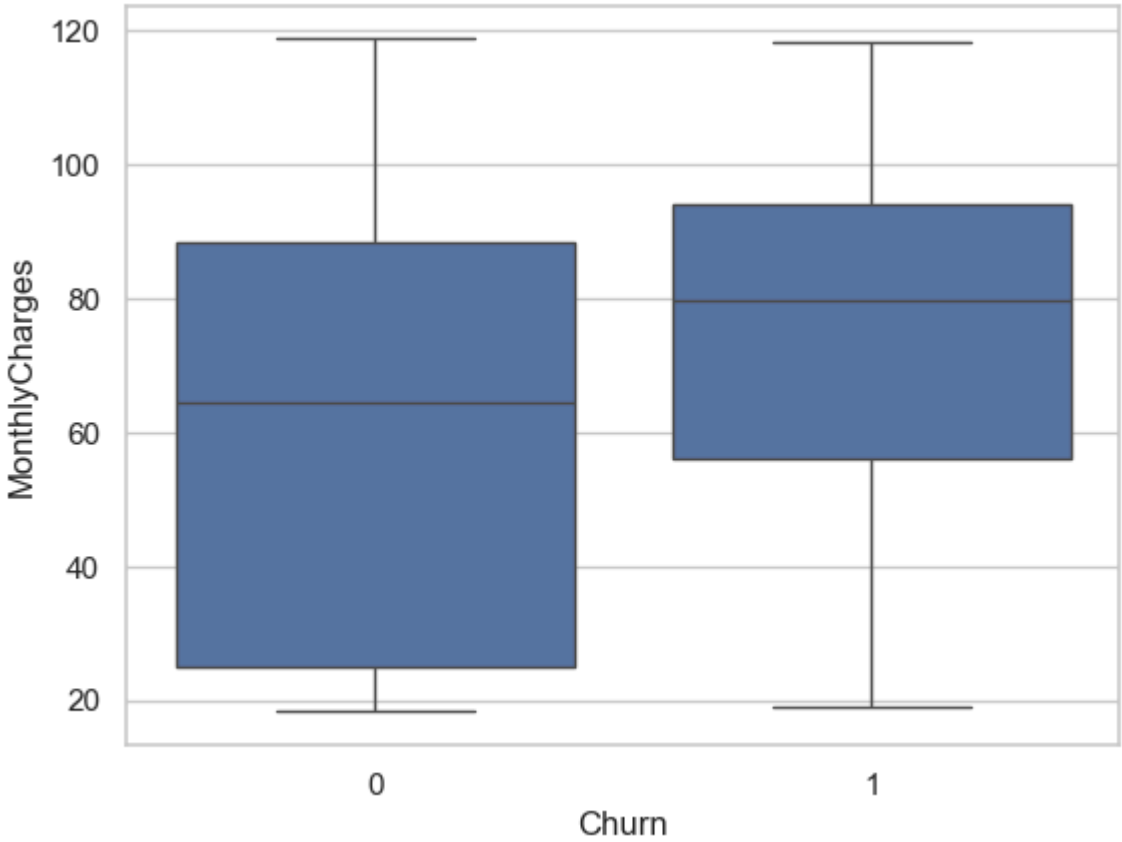
In [ ]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [ ]:
```
churn_dist = df['Churn'].value_counts(normalize=True) * 100

plt.figure(figsize=(6,4))
sns.barplot(x=churn_dist.index, y=churn_dist.values)
plt.xticks([0, 1], ['No Churn', 'Churn'])
plt.ylabel('Percentage (%)')
plt.xlabel('Class')
plt.title('Class Distribution of Churn')
plt.ylim(0, 100)

for i, v in enumerate(churn_dist.values):
    plt.text(i, v + 1, f"{v:.2f}%", ha='center')

plt.tight_layout()
plt.show()
```

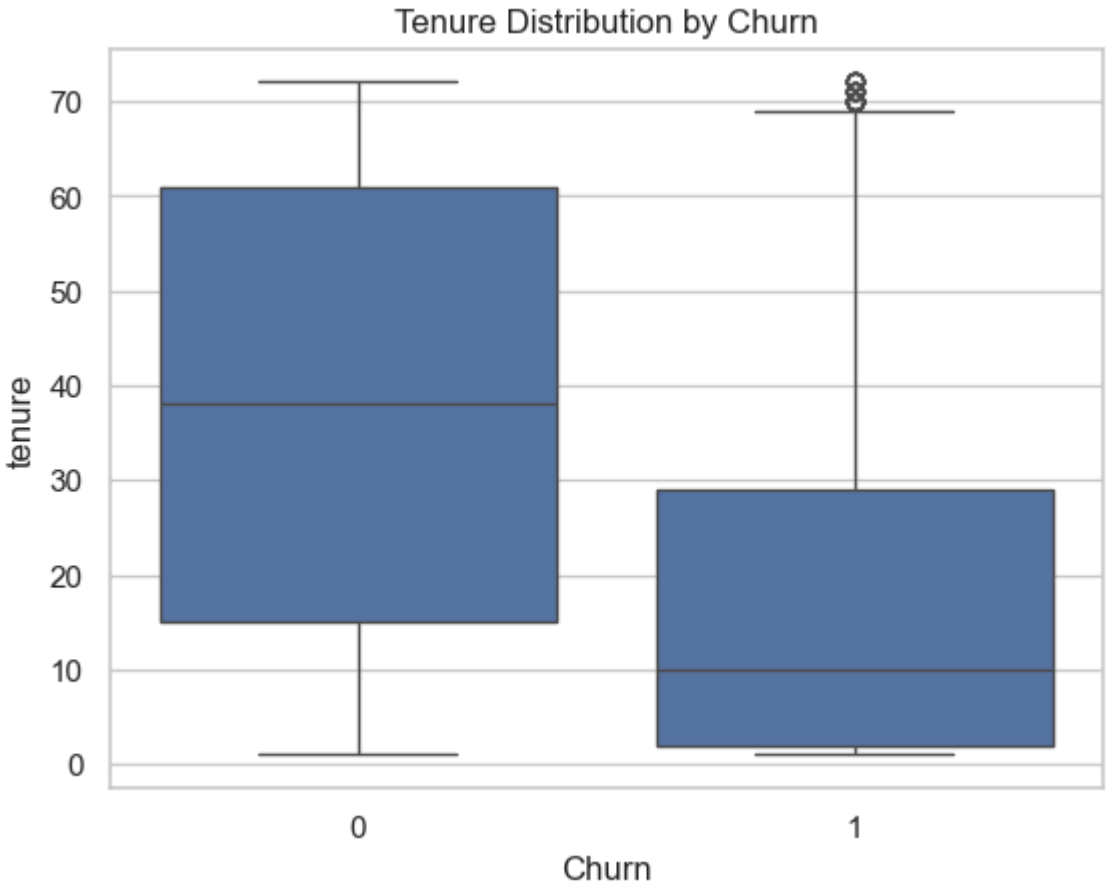


In [ ]:
```
sns.boxplot(x='Churn', y='MonthlyCharges', data=df)
```

Out[ ]:
```
<Axes: xlabel='Churn', ylabel='MonthlyCharges'>
```

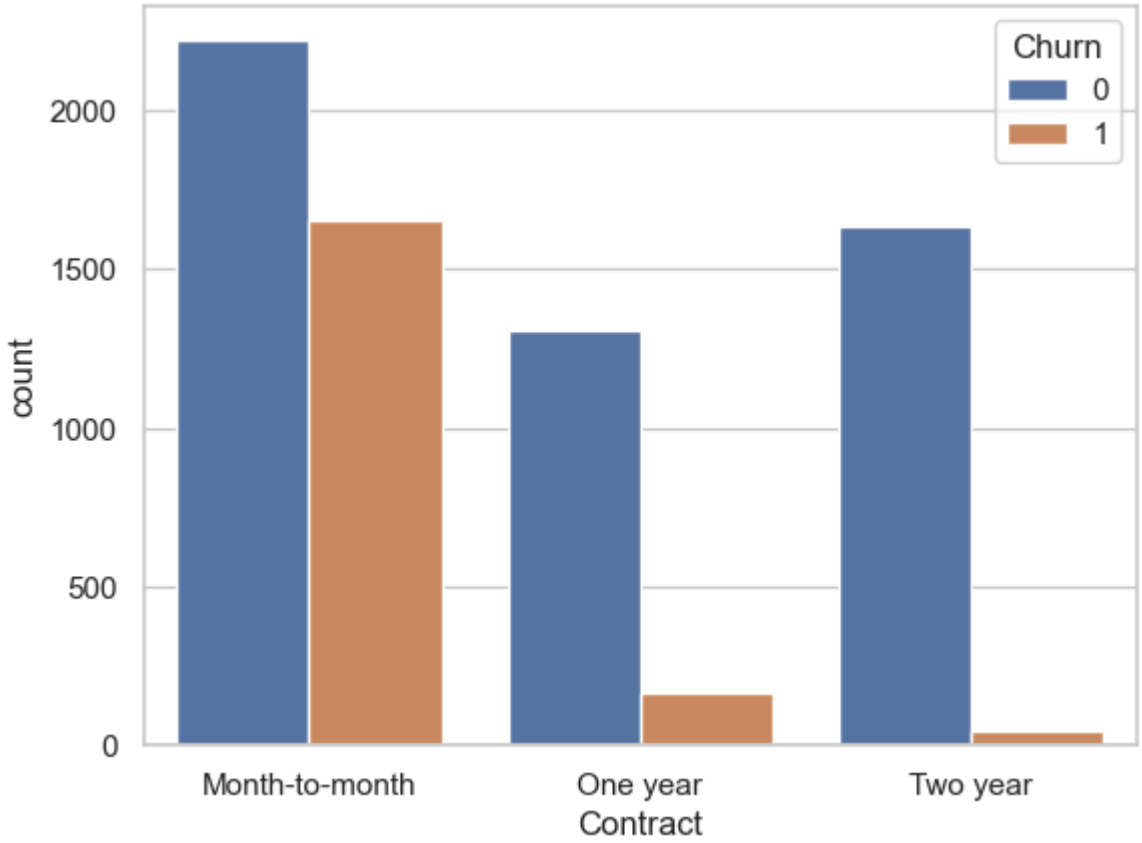


In [ ]:
```
sns.boxplot(x='Churn', y='tenure', data=df)
plt.title("Tenure Distribution by Churn")
plt.show()
```

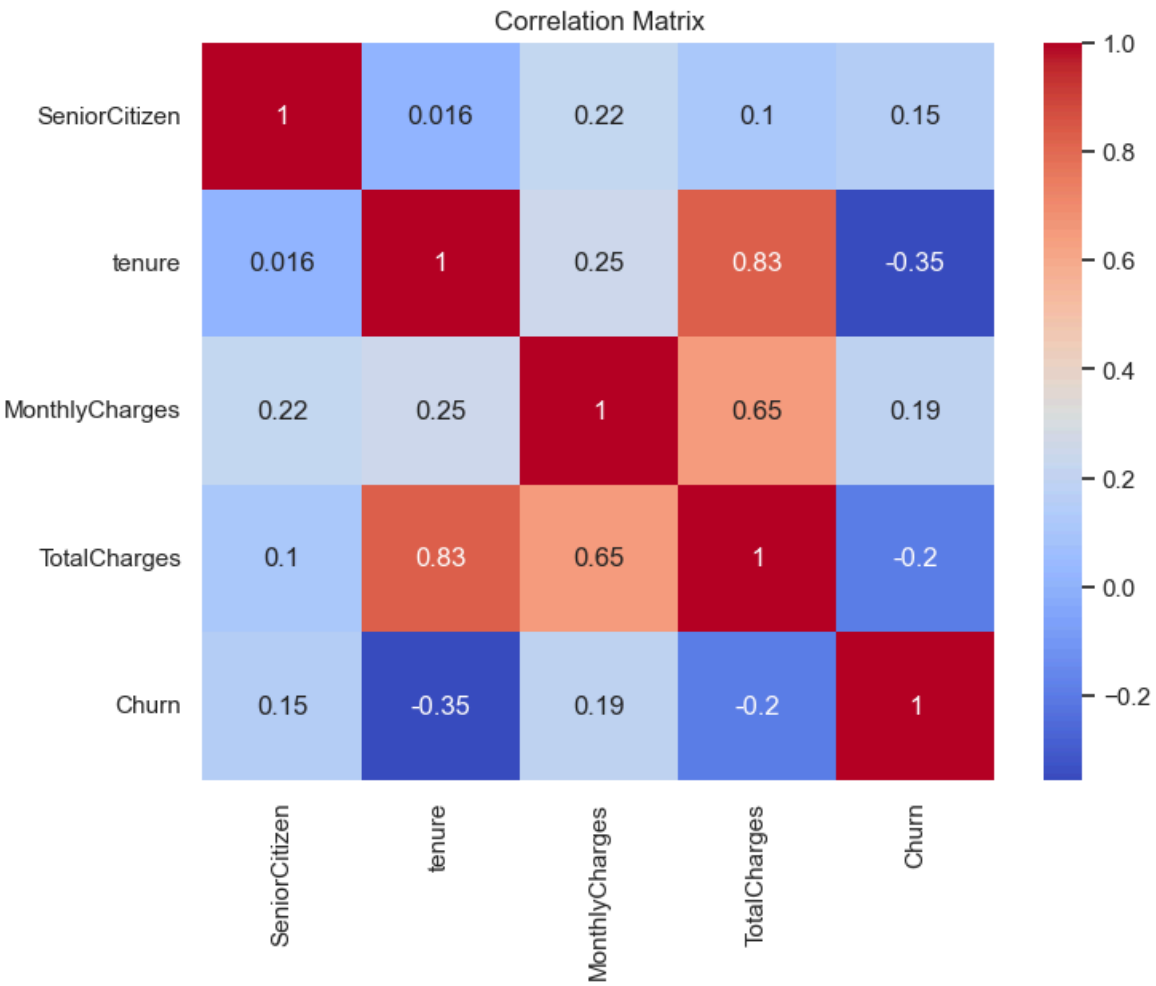


In [ ]:
```
sns.countplot(x='Contract', hue='Churn', data=df)
```

Out[ ]: <Axes: xlabel='Contract', ylabel='count'>



In [ ]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```



In [ ]:
```
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    roc_curve,
    auc,
    ConfusionMatrixDisplay,
)

from sklearn.inspection import permutation_importance
```

In [ ]:
```
RANDOM_STATE = 42
N_ITER_SEARCH = 12
```

In [ ]:
```
import pandas as pd
import numpy as np
```

In [ ]:
```
df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")
print(df.head())
print(df.shape)
```

```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

               PaymentMethod MonthlyCharges  TotalCharges Churn
0           Electronic check          29.85         29.85    No
1               Mailed check          56.95        1889.5    No
2               Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4           Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]
(7043, 21)
```

In [ ]:

```python
data = df.copy()

data.columns = [c.strip() for c in data.columns]

for col in data.select_dtypes(include=["object"]).columns:
    data[col] = data[col].astype(str).str.strip()

data["TotalCharges"] = pd.to_numeric(data["TotalCharges"], errors="coerce")

data["Churn"] = data["Churn"].map({"No": 0, "Yes": 1})

data = data.drop(columns=["customerID"])

print(data.head())
print(data.info())
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  \
0  No phone service             DSL             No          Yes
1               No             DSL            Yes           No
2               No             DSL            Yes          Yes
3  No phone service             DSL            Yes           No
4                No     Fiber optic             No           No

  DeviceProtection TechSupport StreamingTV StreamingMovies        Contract  \
0               No          No          No              No  Month-to-month
1              Yes          No          No              No        One year
2               No          No          No              No  Month-to-month
3              Yes         Yes          No              No        One year
4               No          No          No              No  Month-to-month

  PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges  \
0              Yes           Electronic check           29.85         29.85
1               No               Mailed check           56.95       1889.50
2              Yes               Mailed check           53.85        108.15
3               No  Bank transfer (automatic)           42.30       1840.75
4              Yes           Electronic check           70.70        151.65

   Churn
0      0
1      0
2      1
3      0
4      1
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            7043 non-null   str
 1   SeniorCitizen     7043 non-null   int64
 2   Partner           7043 non-null   str
 3   Dependents        7043 non-null   str
 4   tenure            7043 non-null   int64
 5   PhoneService      7043 non-null   str
 6   MultipleLines     7043 non-null   str
 7   InternetService   7043 non-null   str
 8   OnlineSecurity    7043 non-null   str
 9   OnlineBackup      7043 non-null   str
 10  DeviceProtection  7043 non-null   str
 11  TechSupport       7043 non-null   str
 12  StreamingTV       7043 non-null   str
 13  StreamingMovies   7043 non-null   str
 14  Contract          7043 non-null   str
 15  PaperlessBilling  7043 non-null   str
 16  PaymentMethod     7043 non-null   str
 17  MonthlyCharges    7043 non-null   float64
 18  TotalCharges      7032 non-null   float64
 19  Churn             7043 non-null   int64
dtypes: float64(2), int64(3), str(15)
```

```
memory usage: 1.1 MB
None
```

```
C:\Users\user\AppData\Local\Temp\ipykernel_23324\3421249663.py:5: Pandas4Warning:
For backward compatibility, 'str' dtypes are included by select_dtypes when 'obje
ct' dtype is specified. This behavior is deprecated and will be removed in a futu
re version. Explicitly pass 'str' to `include` to select them, or to `exclude` to
remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-mig
ration-select-dtypes for details on how to write code that works with pandas 2 an
d 3.
  for col in data.select_dtypes(include=["object"]).columns:
```

In [ ]:

```
print(type(data))
print(data.shape)
print(data.columns)
```

```
<class 'pandas.DataFrame'>
(7043, 20)
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
       'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
       'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
       'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
       'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='str')
```

In [ ]:

```
# 1) Average charge per month
tenure_safe = data["tenure"].replace(0, 1)
data["AvgChargesPerMonth"] = data["TotalCharges"] / tenure_safe

# 2) Number of active services
service_cols = [
    "PhoneService",
    "MultipleLines",
    "InternetService",
    "OnlineSecurity",
    "OnlineBackup",
    "DeviceProtection",
    "TechSupport",
    "StreamingTV",
    "StreamingMovies",
]

def service_is_active(x):
    return int(str(x) not in ["No", "No internet service", "No phone service", "

data["NumServices"] = data[service_cols].map(service_is_active).sum(axis=1)

# 3) Long-term customer flag
data["IsLongTermCustomer"] = (data["tenure"] >= 24).astype(int)

# 4) High monthly charges flag
monthly_threshold = data["MonthlyCharges"].median()
data["HighMonthlyCharges"] = (data["MonthlyCharges"] >= monthly_threshold).astyp

# 5) Electronic check flag
data["UsesElectronicCheck"] = (data["PaymentMethod"] == "Electronic check").asty

print(data.head())
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  ...  \
0  No phone service             DSL             No          Yes  ...
1                No             DSL            Yes           No  ...
2                No             DSL            Yes          Yes  ...
3  No phone service             DSL            Yes           No  ...
4                No     Fiber optic             No           No  ...

  PaperlessBilling              PaymentMethod MonthlyCharges TotalCharges  \
0              Yes           Electronic check          29.85        29.85
1               No               Mailed check          56.95      1889.50
2              Yes               Mailed check          53.85       108.15
3               No  Bank transfer (automatic)          42.30      1840.75
4              Yes           Electronic check          70.70       151.65

  Churn AvgChargesPerMonth NumServices  IsLongTermCustomer  \
0     0          29.850000           2                   0
1     0          55.573529           4                   1
2     1          54.075000           4                   0
3     0          40.905556           4                   1
4     1          75.825000           2                   0

   HighMonthlyCharges  UsesElectronicCheck
0                   0                    1
1                   0                    0
2                   0                    0
3                   0                    0
4                   1                    1

[5 rows x 25 columns]
```

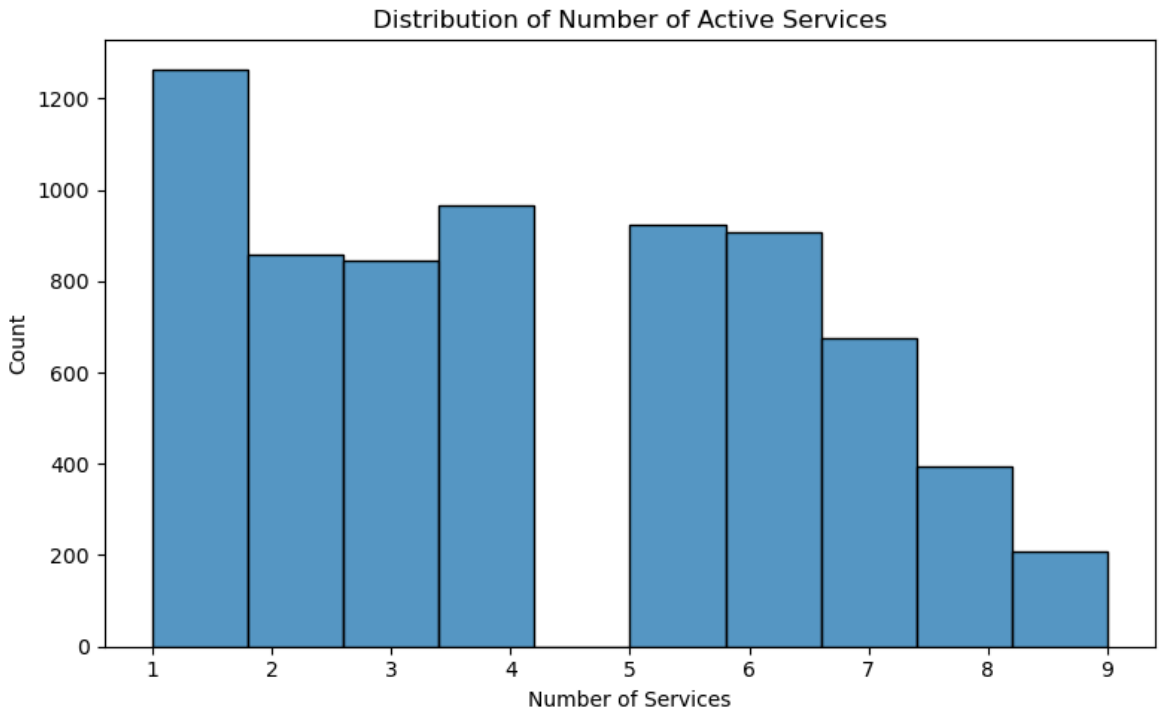
In [ ]:

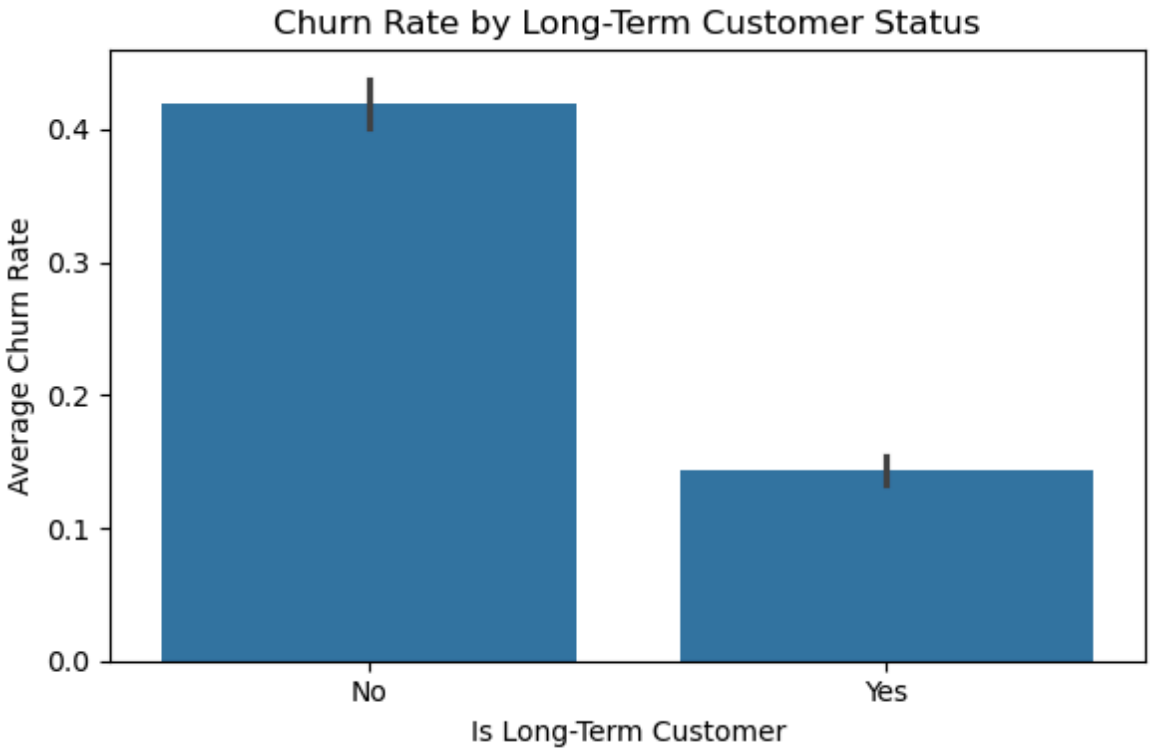
```python
print(data.columns)
print(data[[
    "AvgChargesPerMonth",
    "NumServices",
    "IsLongTermCustomer",
    "HighMonthlyCharges",
    "UsesElectronicCheck"
]].head())
```

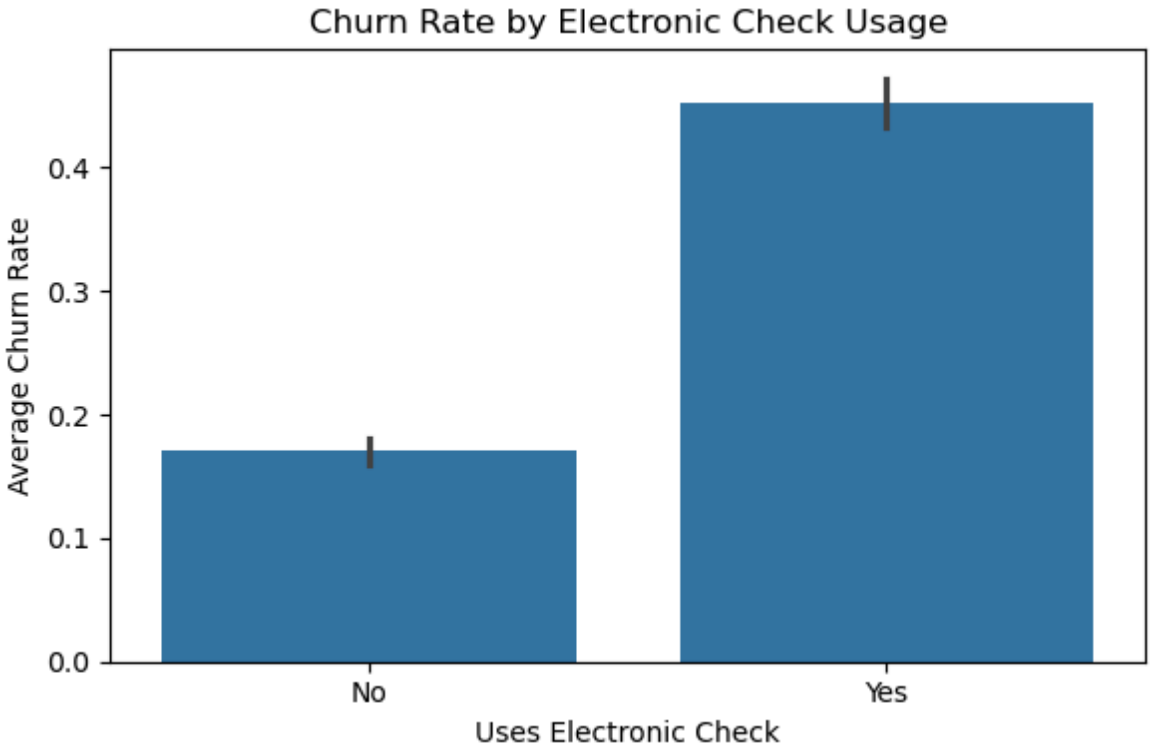
```
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
       'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
       'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
       'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
       'MonthlyCharges', 'TotalCharges', 'Churn', 'AvgChargesPerMonth',
       'NumServices', 'IsLongTermCustomer', 'HighMonthlyCharges',
       'UsesElectronicCheck'],
      dtype='str')
   AvgChargesPerMonth  NumServices  IsLongTermCustomer  HighMonthlyCharges  \
0           29.850000            2                   0                   0
1           55.573529            4                   1                   0
2           54.075000            4                   0                   0
3           40.905556            4                   1                   0
4           75.825000            2                   0                   1

   UsesElectronicCheck
0                    1
1                    0
2                    0
3                    0
4                    1
```

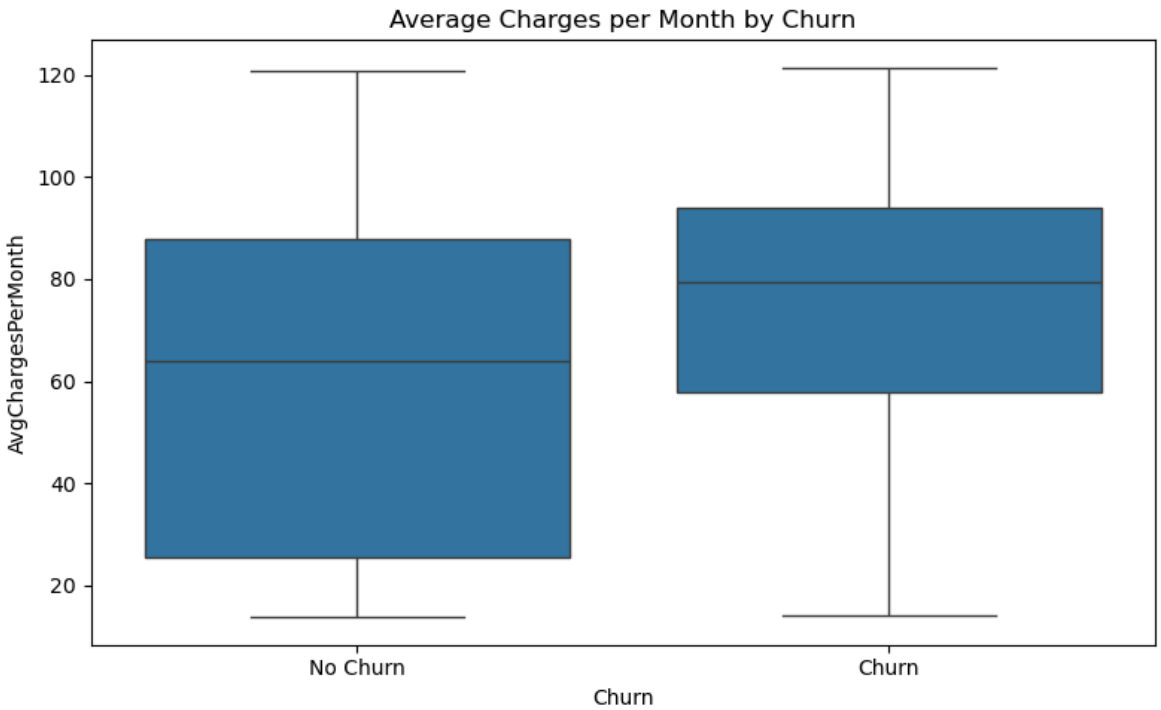
In [ ]:

```
plt.figure(figsize=(8, 5))
sns.histplot(data=data, x="NumServices", bins=10, kde=False)
plt.title("Distribution of Number of Active Services")
plt.xlabel("Number of Services")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 4))
sns.barplot(x="IsLongTermCustomer", y="Churn", data=data)
plt.xticks([0, 1], ["No", "Yes"])
plt.title("Churn Rate by Long-Term Customer Status")
plt.ylabel("Average Churn Rate")
plt.xlabel("Is Long-Term Customer")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 4))
sns.barplot(x="UsesElectronicCheck", y="Churn", data=data)
plt.xticks([0, 1], ["No", "Yes"])
plt.title("Churn Rate by Electronic Check Usage")
plt.ylabel("Average Churn Rate")
plt.xlabel("Uses Electronic Check")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
sns.boxplot(x="Churn", y="AvgChargesPerMonth", data=data)
plt.xticks([0, 1], ["No Churn", "Churn"])
plt.title("Average Charges per Month by Churn")
plt.tight_layout()
plt.show()
```


Distribution of Number of Active Services
Count
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
9
Number of Services


Churn Rate by Long-Term Customer Status
Average Churn Rate
0.0
0.1
0.2
0.3
0.4
No
Yes
Is Long-Term Customer





In [ ]:
```
X = data.drop(columns=["Churn"])
y = data["Churn"].astype(int)

print(X.shape)
print(y.value_counts())
print(X.head())
```

```
(7043, 24)
Churn
0    5174
1    1869
Name: count, dtype: int64
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  ...  \
0  No phone service             DSL             No          Yes  ...
1                No             DSL            Yes           No  ...
2                No             DSL            Yes          Yes  ...
3  No phone service             DSL            Yes           No  ...
4                No     Fiber optic             No           No  ...

         Contract PaperlessBilling              PaymentMethod MonthlyCharges  \
0  Month-to-month              Yes           Electronic check          29.85
1        One year               No               Mailed check          56.95
2  Month-to-month              Yes               Mailed check          53.85
3        One year               No  Bank transfer (automatic)          42.30
4  Month-to-month              Yes           Electronic check          70.70

  TotalCharges AvgChargesPerMonth NumServices  IsLongTermCustomer  \
0        29.85          29.850000           2                   0
1      1889.50          55.573529           4                   1
2       108.15          54.075000           4                   0
3      1840.75          40.905556           4                   1
4       151.65          75.825000           2                   0

   HighMonthlyCharges  UsesElectronicCheck
0                   0                    1
1                   0                    0
2                   0                    0
3                   0                    0
4                   1                    1

[5 rows x 24 columns]
```

In [ ]:
```
print(type(X))
print(type(y))
print(X.shape)
print(y.shape)
```

```
<class 'pandas.DataFrame'>
<class 'pandas.Series'>
(7043, 24)
(7043,)
```

In [ ]:
```
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer

numeric_features = X.select_dtypes(include=["number"]).columns.tolist()
categorical_features = X.select_dtypes(include=["object", "string"]).columns.tol
```

```
print("Numeric features:", numeric_features)
print("Categorical features:", categorical_features)

numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_transformer, numeric_features),
    ("cat", categorical_transformer, categorical_features)
])

print("Preprocessor defined.")
```

```
Numeric features: ['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges',
'AvgChargesPerMonth', 'NumServices', 'IsLongTermCustomer', 'HighMonthlyCharges',
'UsesElectronicCheck']
Categorical features: ['gender', 'Partner', 'Dependents', 'PhoneService', 'Multip
leLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtectio
n', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBillin
g', 'PaymentMethod']
Preprocessor defined.
```

In [ ]:

```
print(type(preprocessor))
print(numeric_features)
print(categorical_features)
```

```
<class 'sklearn.compose._column_transformer.ColumnTransformer'>
['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges', 'AvgChargesPerMont
h', 'NumServices', 'IsLongTermCustomer', 'HighMonthlyCharges', 'UsesElectronicChe
ck']
['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetSer
vice', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'Stre
amingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod']
```

In [ ]:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

print("Train size:", len(X_train))
print("Test size:", len(X_test))
```

```
Train size: 5634
Test size: 1409
```

In [ ]:

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(5634, 24)
(1409, 24)
(5634,)
(1409,)
```

In [ ]:

```
from sklearn.model_selection import GridSearchCV, StratifiedKFold
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from imblearn.pipeline import Pipeline as ImbPipeline
from imblearn.over_sampling import SMOTE
import pandas as pd
import numpy as np

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

models_and_parameters = {
    "LogisticRegression": {
        "model": LogisticRegression(
            max_iter=4000,
            solver="liblinear",
            class_weight="balanced",
            random_state=42
        ),
        "use_smote": False,
        "params": {
            "model__C": [0.01, 0.1, 1, 10]
        }
    },
    "SVM": {
        "model": SVC(
            probability=True,
            class_weight="balanced",
            random_state=42
        ),
        "use_smote": True,
        "params": {
            "model__C": [0.1, 1, 10],
            "model__gamma": ["scale", 0.01, 0.1],
            "model__kernel": ["rbf"]
        }
    },
    "RandomForest": {
        "model": RandomForestClassifier(
            random_state=42,
            class_weight="balanced",
            n_jobs=-1
        ),
        "use_smote": False,
        "params": {
            "model__n_estimators": [200, 300],
            "model__max_depth": [None, 5, 10],
            "model__min_samples_split": [2, 5],
            "model__min_samples_leaf": [1, 2]
        }
    }
}

results = []
```

```
best_estimators = {}

for model_name, config in models_and_parameters.items():
    print(f"\nRunning GridSearchCV for: {model_name}")

    steps = [("preprocess", preprocessor)]
    if config["use_smote"]:
        steps.append(("smote", SMOTE(random_state=42)))
    steps.append(("model", config["model"]))

    pipeline = ImbPipeline(steps=steps)

    grid = GridSearchCV(
        estimator=pipeline,
        param_grid=config["params"],
        scoring="f1",
        cv=cv,
        n_jobs=-1,
        verbose=1,
        refit=True
    )

    grid.fit(X_train, y_train)

    best_model = grid.best_estimator_
    best_estimators[model_name] = best_model

    y_pred = best_model.predict(X_test)

    if hasattr(best_model, "predict_proba"):
        y_proba = best_model.predict_proba(X_test)[:, 1]
        roc_auc = roc_auc_score(y_test, y_proba)
    else:
        roc_auc = np.nan

    results.append({
        "Model": model_name,
        "Best_Params": grid.best_params_,
        "Best_CV_F1": grid.best_score_,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred, zero_division=0),
        "Recall": recall_score(y_test, y_pred, zero_division=0),
        "F1": f1_score(y_test, y_pred, zero_division=0),
        "ROC_AUC": roc_auc
    })

results_df = pd.DataFrame(results).sort_values(by="F1", ascending=False)
results_df
```

```
Running GridSearchCV for: LogisticRegression
Fitting 5 folds for each of 4 candidates, totalling 20 fits

Running GridSearchCV for: SVM
Fitting 5 folds for each of 9 candidates, totalling 45 fits

Running GridSearchCV for: RandomForest
Fitting 5 folds for each of 24 candidates, totalling 120 fits
```

Out[ ]:

	Model	Best_Params	Best_CV_F1	Accuracy	Precision	Recall
2	RandomForest	{'model__max_depth': 10, 'model__min_samples_l...	0.632392	0.770759	0.552577	0.716578
1	SVM	{'model__C': 10, 'model__gamma': 0.01, 'model_...	0.630806	0.756565	0.529190	0.751337
0	LogisticRegression	{'model__C': 1}	0.631331	0.737402	0.503436	0.783422

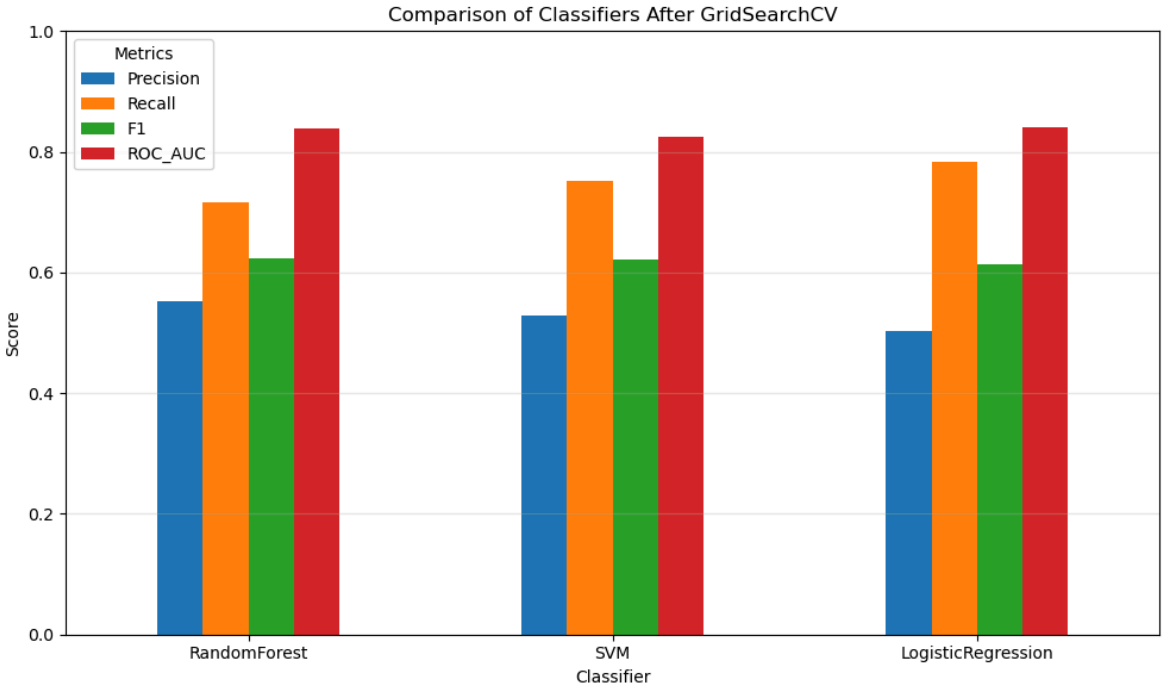
In [ ]:

```
print(type(results_df))
print(results_df.head())
print(results_df.iloc[0])
```

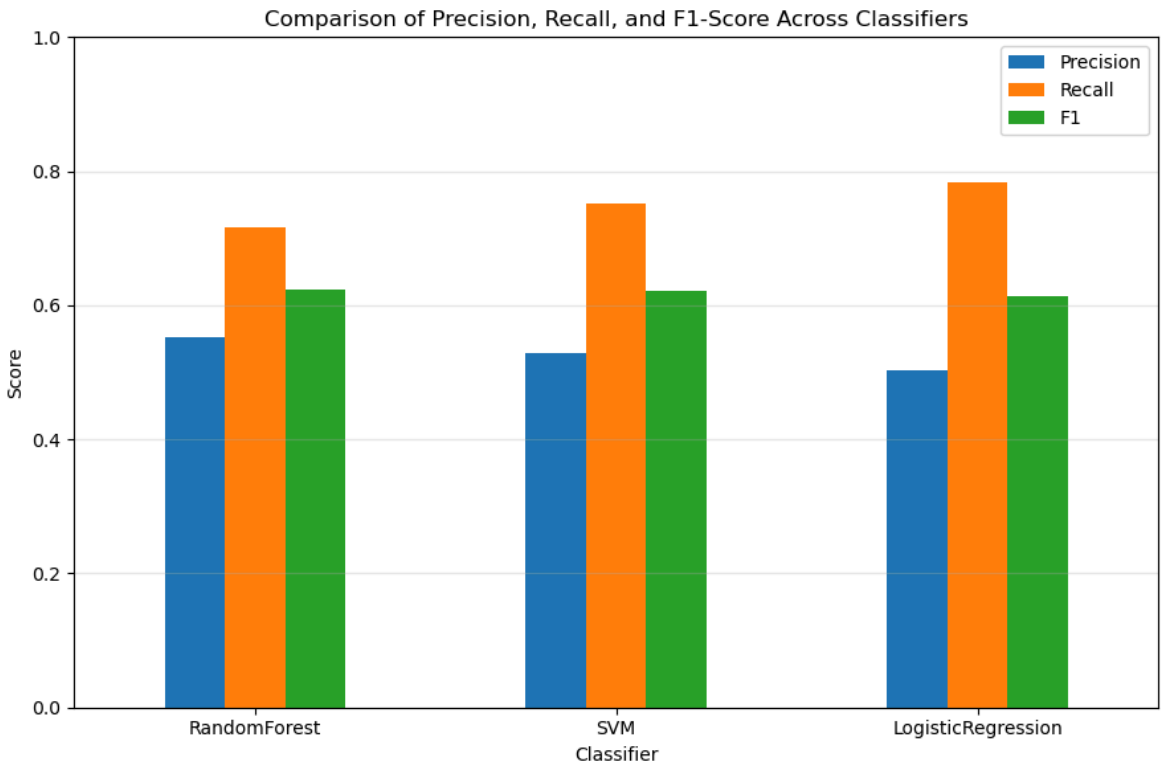
```
<class 'pandas.DataFrame'>
                Model                                        Best_Params  \
2        RandomForest  {'model__max_depth': 10, 'model__min_samples_l...
1                 SVM  {'model__C': 10, 'model__gamma': 0.01, 'model_...
0  LogisticRegression                                      {'model__C': 1}

   Best_CV_F1  Accuracy  Precision    Recall        F1   ROC_AUC
2    0.632392  0.770759   0.552577  0.716578  0.623981  0.839265
1    0.630806  0.756565   0.529190  0.751337  0.620994  0.825811
0    0.631331  0.737402   0.503436  0.783422  0.612971  0.840562
Model                                                RandomForest
Best_Params    {'model__max_depth': 10, 'model__min_samples_l...
Best_CV_F1                                              0.632392
Accuracy                                                0.770759
Precision                                               0.552577
Recall                                                  0.716578
F1                                                      0.623981
ROC_AUC                                                 0.839265
Name: 2, dtype: object
```

In [ ]:

```
import matplotlib.pyplot as plt
import pandas as pd

plot_df = results_df.copy()

metrics_to_plot = ["Precision", "Recall", "F1", "ROC_AUC"]
plot_df = plot_df[["Model"] + metrics_to_plot].set_index("Model")

ax = plot_df.plot(kind="bar", figsize=(10, 6))
plt.title("Comparison of Classifiers After GridSearchCV")
plt.ylabel("Score")
plt.xlabel("Classifier")
plt.xticks(rotation=0)
plt.ylim(0, 1)
plt.legend(title="Metrics")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```



In [ ]:
```
plot_df = results_df[["Model", "Precision", "Recall", "F1"]].set_index("Model")

ax = plot_df.plot(kind="bar", figsize=(9, 6))
plt.title("Comparison of Precision, Recall, and F1-Score Across Classifiers")
plt.ylabel("Score")
plt.xlabel("Classifier")
plt.xticks(rotation=0)
plt.ylim(0, 1)
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```



In [ ]:
```
tenure_safe = data["tenure"].replace(0, 1)
data["AvgChargesPerMonth"] = data["TotalCharges"] / tenure_safe
```

```
service_cols = [
    "PhoneService",
    "MultipleLines",
    "InternetService",
    "OnlineSecurity",
    "OnlineBackup",
    "DeviceProtection",
    "TechSupport",
    "StreamingTV",
    "StreamingMovies",
]

def service_is_active(x):
    return int(str(x) not in ["No", "No internet service", "No phone service", "

data["NumServices"] = data[service_cols].map(service_is_active).sum(axis=1)

data["IsLongTermCustomer"] = (data["tenure"] >= 24).astype(int)

monthly_threshold = data["MonthlyCharges"].median()
data["HighMonthlyCharges"] = (data["MonthlyCharges"] >= monthly_threshold).astyp

data["UsesElectronicCheck"] = (data["PaymentMethod"] == "Electronic check").asty

print(data.head())
print(data.columns)
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  \
0  No phone service             DSL             No          Yes
1                No             DSL            Yes           No
2                No             DSL            Yes          Yes
3  No phone service             DSL            Yes           No
4                No     Fiber optic             No           No

  DeviceProtection TechSupport StreamingTV StreamingMovies        Contract  \
0               No          No          No              No  Month-to-month
1              Yes          No          No              No        One year
2               No          No          No              No  Month-to-month
3              Yes         Yes          No              No        One year
4               No          No          No              No  Month-to-month

  PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges  \
0              Yes           Electronic check           29.85         29.85
1               No               Mailed check           56.95       1889.50
2              Yes               Mailed check           53.85        108.15
3               No  Bank transfer (automatic)           42.30       1840.75
4              Yes           Electronic check           70.70        151.65

   Churn  AvgChargesPerMonth  NumServices  IsLongTermCustomer  \
0      0           29.850000            2                   0
1      0           55.573529            4                   1
2      1           54.075000            4                   0
3      0           40.905556            4                   1
4      1           75.825000            2                   0

   HighMonthlyCharges  UsesElectronicCheck
0                   0                    1
1                   0                    0
2                   0                    0
3                   0                    0
4                   1                    1
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
       'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
       'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
       'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
       'MonthlyCharges', 'TotalCharges', 'Churn', 'AvgChargesPerMonth',
       'NumServices', 'IsLongTermCustomer', 'HighMonthlyCharges',
       'UsesElectronicCheck'],
      dtype='str')
```

In [ ]:

```python
data = df.copy()

data.columns = [c.strip() for c in data.columns]

for col in data.select_dtypes(include=["object", "string"]).columns:
    data[col] = data[col].astype(str).str.strip()

data["TotalCharges"] = pd.to_numeric(data["TotalCharges"], errors="coerce")
data["Churn"] = data["Churn"].map({"No": 0, "Yes": 1})
data = data.drop(columns=["customerID"])
```

```
print(data.head())
print(data.shape)
print(data.dtypes)
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  \
0  No phone service             DSL             No          Yes
1               No             DSL            Yes           No
2               No             DSL            Yes          Yes
3  No phone service             DSL            Yes           No
4               No     Fiber optic             No           No

  DeviceProtection TechSupport StreamingTV StreamingMovies        Contract  \
0               No          No          No              No  Month-to-month
1              Yes          No          No              No        One year
2               No          No          No              No  Month-to-month
3              Yes         Yes          No              No        One year
4               No          No          No              No  Month-to-month

  PaperlessBilling              PaymentMethod  MonthlyCharges  TotalCharges  \
0              Yes           Electronic check           29.85         29.85
1               No               Mailed check           56.95       1889.50
2              Yes               Mailed check           53.85        108.15
3               No  Bank transfer (automatic)           42.30       1840.75
4              Yes           Electronic check           70.70        151.65

   Churn
0      0
1      0
2      1
3      0
4      1
(7043, 20)
gender                  str
SeniorCitizen         int64
Partner                 str
Dependents              str
tenure                int64
PhoneService            str
MultipleLines           str
InternetService         str
OnlineSecurity          str
OnlineBackup            str
DeviceProtection        str
TechSupport             str
StreamingTV             str
StreamingMovies         str
Contract                str
PaperlessBilling        str
PaymentMethod           str
MonthlyCharges      float64
TotalCharges        float64
Churn                 int64
dtype: object
```

In [ ]:

```python
# 1) Average charge per month
tenure_safe = data["tenure"].replace(0, 1)
data["AvgChargesPerMonth"] = data["TotalCharges"] / tenure_safe
```

```
# 2) Number of active services
service_cols = [
    "PhoneService",
    "MultipleLines",
    "InternetService",
    "OnlineSecurity",
    "OnlineBackup",
    "DeviceProtection",
    "TechSupport",
    "StreamingTV",
    "StreamingMovies",
]

def service_is_active(x):
    return int(str(x) not in ["No", "No internet service", "No phone service", "

data["NumServices"] = data[service_cols].map(service_is_active).sum(axis=1)

# 3) Long-term customer flag
data["IsLongTermCustomer"] = (data["tenure"] >= 24).astype(int)

# 4) High monthly charges flag
monthly_threshold = data["MonthlyCharges"].median()
data["HighMonthlyCharges"] = (data["MonthlyCharges"] >= monthly_threshold).astyp

# 5) Electronic check flag
data["UsesElectronicCheck"] = (data["PaymentMethod"] == "Electronic check").asty

print(data.head())
```

```
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  ...  \
0  No phone service             DSL             No          Yes  ...
1                No             DSL            Yes           No  ...
2                No             DSL            Yes          Yes  ...
3  No phone service             DSL            Yes           No  ...
4                No     Fiber optic             No           No  ...

  PaperlessBilling              PaymentMethod MonthlyCharges TotalCharges  \
0              Yes           Electronic check          29.85        29.85
1               No               Mailed check          56.95      1889.50
2              Yes               Mailed check          53.85       108.15
3               No  Bank transfer (automatic)          42.30      1840.75
4              Yes           Electronic check          70.70       151.65

  Churn AvgChargesPerMonth NumServices  IsLongTermCustomer  \
0     0          29.850000           2                   0
1     0          55.573529           4                   1
2     1          54.075000           4                   0
3     0          40.905556           4                   1
4     1          75.825000           2                   0

   HighMonthlyCharges  UsesElectronicCheck
0                   0                    1
1                   0                    0
2                   0                    0
3                   0                    0
4                   1                    1

[5 rows x 25 columns]
```

In [ ]:

```python
X = data.drop(columns=["Churn"])
y = data["Churn"].astype(int)

print(X.shape)
print(y.value_counts())
print(X.head())
```

```
(7043, 24)
Churn
0    5174
1    1869
Name: count, dtype: int64
   gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  Female              0     Yes         No       1           No
1    Male              0      No         No      34          Yes
2    Male              0      No         No       2          Yes
3    Male              0      No         No      45           No
4  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity OnlineBackup  ...  \
0  No phone service             DSL             No          Yes  ...
1                No             DSL            Yes           No  ...
2                No             DSL            Yes          Yes  ...
3  No phone service             DSL            Yes           No  ...
4                No     Fiber optic             No           No  ...

         Contract PaperlessBilling              PaymentMethod MonthlyCharges  \
0  Month-to-month              Yes           Electronic check          29.85
1        One year               No               Mailed check          56.95
2  Month-to-month              Yes               Mailed check          53.85
3        One year               No  Bank transfer (automatic)          42.30
4  Month-to-month              Yes           Electronic check          70.70

  TotalCharges AvgChargesPerMonth NumServices  IsLongTermCustomer  \
0        29.85          29.850000           2                   0
1      1889.50          55.573529           4                   1
2       108.15          54.075000           4                   0
3      1840.75          40.905556           4                   1
4       151.65          75.825000           2                   0

   HighMonthlyCharges  UsesElectronicCheck
0                   0                    1
1                   0                    0
2                   0                    0
3                   0                    0
4                   1                    1

[5 rows x 24 columns]
```

In [ ]:

```
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer

numeric_features = X.select_dtypes(include=["number"]).columns.tolist()
categorical_features = X.select_dtypes(include=["object", "string"]).columns.tol

print("Numeric features:", numeric_features)
print("Categorical features:", categorical_features)

numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
```

```
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_transformer, numeric_features),
    ("cat", categorical_transformer, categorical_features)
])

print("Preprocessor defined.")
```

```
Numeric features: ['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges',
'AvgChargesPerMonth', 'NumServices', 'IsLongTermCustomer', 'HighMonthlyCharges',
'UsesElectronicCheck']
Categorical features: ['gender', 'Partner', 'Dependents', 'PhoneService', 'Multip
leLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtectio
n', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBillin
g', 'PaymentMethod']
Preprocessor defined.
```

In [ ]:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

print("Train size:", len(X_train))
print("Test size:", len(X_test))
```

```
Train size: 5634
Test size: 1409
```

In [ ]:

```
from sklearn.model_selection import GridSearchCV, StratifiedKFold
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from imblearn.pipeline import Pipeline as ImbPipeline
from imblearn.over_sampling import SMOTE
import pandas as pd
import numpy as np

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

models_and_parameters = {
    "LogisticRegression": {
        "model": LogisticRegression(
            max_iter=4000,
            solver="liblinear",
            class_weight="balanced",
            random_state=42
        ),
        "use_smote": False,
        "params": {
            "model__C": [0.01, 0.1, 1, 10]
        }
    },
    "SVM": {
        "model": SVC(
            probability=True,
            class_weight="balanced",
            random_state=42
```

```
        ),
        "use_smote": True,
        "params": {
            "model__C": [0.1, 1, 10],
            "model__gamma": ["scale", 0.01, 0.1],
            "model__kernel": ["rbf"]
        }
    },
    "RandomForest": {
        "model": RandomForestClassifier(
            random_state=42,
            class_weight="balanced",
            n_jobs=-1
        ),
        "use_smote": False,
        "params": {
            "model__n_estimators": [200, 300],
            "model__max_depth": [None, 5, 10],
            "model__min_samples_split": [2, 5],
            "model__min_samples_leaf": [1, 2]
        }
    }
}

results = []
best_estimators = {}

for model_name, config in models_and_parameters.items():
    print(f"\nRunning GridSearchCV for: {model_name}")

    steps = [("preprocess", preprocessor)]
    if config["use_smote"]:
        steps.append(("smote", SMOTE(random_state=42)))
    steps.append(("model", config["model"]))

    pipeline = ImbPipeline(steps=steps)

    grid = GridSearchCV(
        estimator=pipeline,
        param_grid=config["params"],
        scoring="f1",
        cv=cv,
        n_jobs=-1,
        verbose=1,
        refit=True
    )

    grid.fit(X_train, y_train)

    best_model = grid.best_estimator_
    best_estimators[model_name] = best_model

    y_pred = best_model.predict(X_test)

    if hasattr(best_model, "predict_proba"):
        y_proba = best_model.predict_proba(X_test)[:, 1]
        roc_auc = roc_auc_score(y_test, y_proba)
    else:
        roc_auc = np.nan
```

```
    results.append({
        "Model": model_name,
        "Best_Params": grid.best_params_,
        "Best_CV_F1": grid.best_score_,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred, zero_division=0),
        "Recall": recall_score(y_test, y_pred, zero_division=0),
        "F1": f1_score(y_test, y_pred, zero_division=0),
        "ROC_AUC": roc_auc
    })

results_df = pd.DataFrame(results).sort_values(by="F1", ascending=False)
results_df
```

```
Running GridSearchCV for: LogisticRegression
Fitting 5 folds for each of 4 candidates, totalling 20 fits

Running GridSearchCV for: SVM
Fitting 5 folds for each of 9 candidates, totalling 45 fits

Running GridSearchCV for: RandomForest
Fitting 5 folds for each of 24 candidates, totalling 120 fits
```

Out[ ]:

	Model	Best_Params	Best_CV_F1	Accuracy	Precision	Recall
2	RandomForest	{'model__max_depth': 10, 'model__min_samples_l...	0.632392	0.770759	0.552577	0.716578
1	SVM	{'model__C': 10, 'model__gamma': 0.01, 'model_...	0.630806	0.756565	0.529190	0.751337
0	LogisticRegression	{'model__C': 1}	0.631331	0.737402	0.503436	0.783422

In [ ]:

```
numeric_cols = X.select_dtypes(include=["number"]).columns.tolist()
categorical_cols = X.select_dtypes(exclude=["number"]).columns.tolist()

numeric_pipeline = Pipeline(
    steps=[
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()),
    ]
)

categorical_pipeline = Pipeline(
    steps=[
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore", sparse_output=False)),
    ]
)

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_pipeline, numeric_cols),
        ("cat", categorical_pipeline, categorical_cols),
    ]
)
```

```
print("Numeric columns:", len(numeric_cols))
print("Categorical columns:", len(categorical_cols))
```

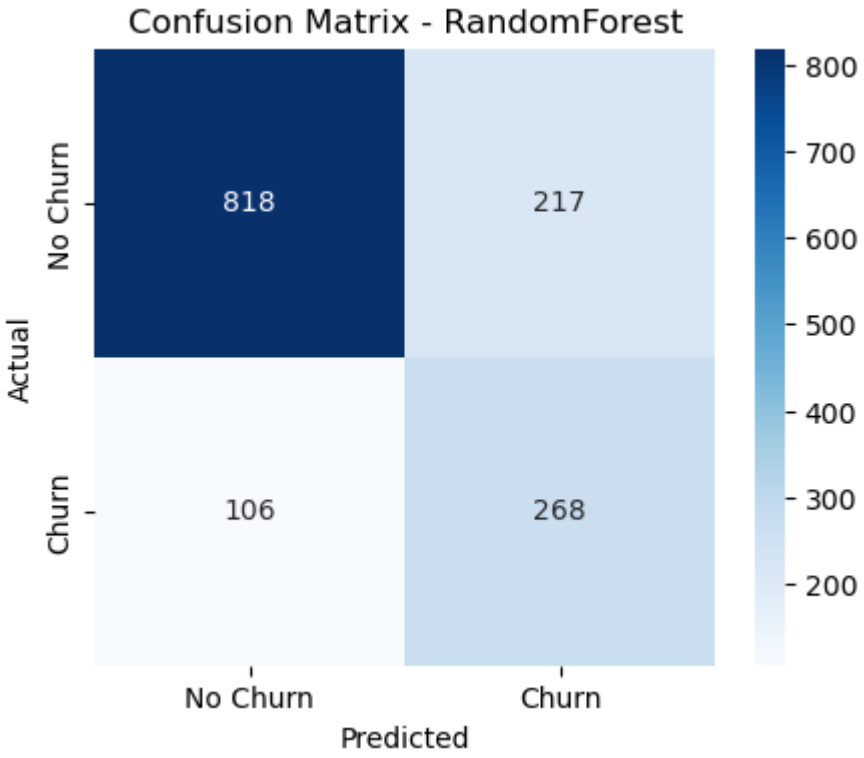
```
Numeric columns: 9
Categorical columns: 15
```

In [ ]:
```
print(results_df)
print("\nBest model:")
print(results_df.iloc[0])
```

```
                Model                                        Best_Params  \
2        RandomForest  {'model__max_depth': 10, 'model__min_samples_l...
1                 SVM  {'model__C': 10, 'model__gamma': 0.01, 'model_...
0  LogisticRegression                                     {'model__C': 1}

   Best_CV_F1  Accuracy  Precision    Recall        F1   ROC_AUC
2    0.632392  0.770759   0.552577  0.716578  0.623981  0.839265
1    0.630806  0.756565   0.529190  0.751337  0.620994  0.825811
0    0.631331  0.737402   0.503436  0.783422  0.612971  0.840562

Best model:
Model                                                RandomForest
Best_Params    {'model__max_depth': 10, 'model__min_samples_l...
Best_CV_F1                                                0.632392
Accuracy                                                  0.770759
Precision                                                 0.552577
Recall                                                    0.716578
F1                                                        0.623981
ROC_AUC                                                   0.839265
Name: 2, dtype: object
```

In [ ]:

In [ ]:
```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

best_model_name = results_df.iloc[0]["Model"]
best_model = best_estimators[best_model_name]

y_pred_best = best_model.predict(X_test)
cm = confusion_matrix(y_test, y_pred_best)

plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["No Churn", "Churn"],
            yticklabels=["No Churn", "Churn"])
plt.title(f"Confusion Matrix - {best_model_name}")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



In [ ]:

In [ ]:
```
X_temp, X_test, y_temp, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=RANDOM_STATE,
    stratify=y,
)

X_train, X_valid, y_train, y_valid = train_test_split(
    X_temp,
    y_temp,
    test_size=0.25,
    random_state=RANDOM_STATE,
    stratify=y_temp,
)

print("Train size:", len(X_train))
print("Validation size:", len(X_valid))
print("Test size:", len(X_test))
```

```
Train size: 4225
Validation size: 1409
Test size: 1409
```

In [ ]:
```
def find_best_threshold(y_true, y_proba, objective="f1"):
    thresholds = np.arange(0.20, 0.81, 0.02)
    best_threshold = 0.50
    best_score = -1
    best_metrics = {}

    for threshold in thresholds:
        y_pred = (y_proba >= threshold).astype(int)

        prec = precision_score(y_true, y_pred, zero_division=0)
```

```
        rec = recall_score(y_true, y_pred, zero_division=0)
        f1 = f1_score(y_true, y_pred, zero_division=0)

        if objective == "precision":
            score = prec
        elif objective == "recall":
            score = rec
        else:
            score = f1

        if score > best_score:
            best_score = score
            best_threshold = threshold
            best_metrics = {
                "Precision_valid": prec,
                "Recall_valid": rec,
                "F1_valid": f1,
            }

    return best_threshold, best_metrics
```

In [ ]:
```
RANDOM_STATE = 42
N_ITER_SEARCH = 10
```

In [ ]:
```
try:
    from xgboost import XGBClassifier
    xgb_available = True
except ImportError:
    XGBClassifier = None
    xgb_available = False

print("XGBoost available:", xgb_available)
```

```
XGBoost available: True
```

In [ ]:
```
pos = (y_train == 1).sum()
neg = (y_train == 0).sum()
scale_pos_weight = neg / pos

model_configs = {
"LogisticRegression": {
    "model": LogisticRegression(
        max_iter=4000,
        solver="liblinear",
        class_weight="balanced",
        random_state=RANDOM_STATE,
    ),
    "use_smote": False,
    "params": {
        "model__C": np.logspace(-2, 2, 10),
    },
},
    "SVM": {
        "model": SVC(
            probability=True,
            class_weight="balanced",
            random_state=RANDOM_STATE,
        ),
        "use_smote": True,
        "params": {
```

```
            "model__C": [0.1, 0.5, 1, 2, 5, 10],
            "model__gamma": ["scale", 0.01, 0.05, 0.1, 0.5],
            "model__kernel": ["rbf"],
        },
    },
    "RandomForest": {
        "model": RandomForestClassifier(
            random_state=RANDOM_STATE,
            n_jobs=-1,
            class_weight="balanced",
        ),
        "use_smote": False,
        "params": {
            "model__n_estimators": [200, 300, 500, 700],
            "model__max_depth": [None, 5, 8, 12, 16],
            "model__min_samples_split": [2, 5, 10, 20],
            "model__min_samples_leaf": [1, 2, 4, 8],
        },
    },
}

if xgb_available:
    model_configs["XGBoost"] = {
        "model": XGBClassifier(
            random_state=RANDOM_STATE,
            eval_metric="logloss",
            n_jobs=-1,
            scale_pos_weight=scale_pos_weight,
        ),
        "use_smote": False,
        "params": {
            "model__n_estimators": [200, 300, 500, 700],
            "model__max_depth": [3, 4, 5, 6, 8],
            "model__learning_rate": [0.01, 0.03, 0.05, 0.1],
            "model__subsample": [0.7, 0.8, 0.9, 1.0],
            "model__colsample_bytree": [0.7, 0.8, 0.9, 1.0],
            "model__min_child_weight": [1, 3, 5],
        },
    }

print(model_configs.keys())
```

```
dict_keys(['LogisticRegression', 'SVM', 'RandomForest', 'XGBoost'])
```

In [ ]:

```
pos = (y_train == 1).sum()
neg = (y_train == 0).sum()
scale_pos_weight = neg / pos

model_configs = {
    "LogisticRegression": {
        "model": LogisticRegression(
            max_iter=4000,
            solver="liblinear",
            class_weight="balanced",
            random_state=RANDOM_STATE,
        ),
        "use_smote": False,
        "params": {
            "model__C": np.logspace(-2, 2, 10),
        },
```

```
    },
    "SVM": {
        "model": SVC(
            probability=True,
            class_weight="balanced",
            random_state=RANDOM_STATE,
        ),
        "use_smote": True,
        "params": {
            "model__C": [0.1, 0.5, 1, 2, 5, 10],
            "model__gamma": ["scale", 0.01, 0.05, 0.1, 0.5],
            "model__kernel": ["rbf"],
        },
    },
    "RandomForest": {
        "model": RandomForestClassifier(
            random_state=RANDOM_STATE,
            n_jobs=-1,
            class_weight="balanced",
        ),
        "use_smote": False,
        "params": {
            "model__n_estimators": [200, 300, 500, 700],
            "model__max_depth": [None, 5, 8, 12, 16],
            "model__min_samples_split": [2, 5, 10, 20],
            "model__min_samples_leaf": [1, 2, 4, 8],
        },
    },
}

if xgb_available:
    model_configs["XGBoost"] = {
        "model": XGBClassifier(
            random_state=RANDOM_STATE,
            eval_metric="logloss",
            n_jobs=-1,
            scale_pos_weight=scale_pos_weight,
        ),
        "use_smote": False,
        "params": {
            "model__n_estimators": [200, 300, 500, 700],
            "model__max_depth": [3, 4, 5, 6, 8],
            "model__learning_rate": [0.01, 0.03, 0.05, 0.1],
            "model__subsample": [0.7, 0.8, 0.9, 1.0],
            "model__colsample_bytree": [0.7, 0.8, 0.9, 1.0],
            "model__min_child_weight": [1, 3, 5],
        },
    }
```

In [ ]:

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)

results = []
artifacts = {}
trained_pipelines = {}

threshold_objective = "f1"   # change to "recall" or "precision" if needed

for model_name, config in model_configs.items():
    print(f"\nTraining and tuning: {model_name}")
```

```
    steps = [("preprocess", preprocessor)]
    if config["use_smote"]:
        steps.append(("smote", SMOTE(random_state=RANDOM_STATE)))
    steps.append(("model", config["model"]))

    pipeline = ImbPipeline(steps=steps)

    search = RandomizedSearchCV(
        estimator=pipeline,
        param_distributions=config["params"],
        n_iter=N_ITER_SEARCH,
        scoring="f1",
        cv=cv,
        verbose=1,
        n_jobs=-1,
        random_state=RANDOM_STATE,
        refit=True,
    )

    search.fit(X_train, y_train)
    best_pipeline = search.best_estimator_
    trained_pipelines[model_name] = best_pipeline

    # Threshold tuning on validation set
    y_valid_proba = best_pipeline.predict_proba(X_valid)[:, 1]
    best_threshold, valid_metrics = find_best_threshold(
        y_valid, y_valid_proba, objective=threshold_objective
    )

    # Final evaluation on test set
    y_test_proba = best_pipeline.predict_proba(X_test)[:, 1]
    y_test_pred = (y_test_proba >= best_threshold).astype(int)

    acc = accuracy_score(y_test, y_test_pred)
    prec = precision_score(y_test, y_test_pred, zero_division=0)
    rec = recall_score(y_test, y_test_pred, zero_division=0)
    f1 = f1_score(y_test, y_test_pred, zero_division=0)

    fpr, tpr, _ = roc_curve(y_test, y_test_proba)
    roc_auc = auc(fpr, tpr)
    cm = confusion_matrix(y_test, y_test_pred)

    results.append({
        "Model": model_name,
        "BestCV_F1": search.best_score_,
        "Threshold": best_threshold,
        "Accuracy": acc,
        "Precision": prec,
        "Recall": rec,
        "F1": f1,
        "ROC_AUC": roc_auc,
        "BestParams": search.best_params_,
    })

    artifacts[model_name] = {
        "confusion_matrix": cm,
        "fpr": fpr,
        "tpr": tpr,
    }
```

```
results_df = pd.DataFrame(results).sort_values(by="F1", ascending=False)
results_df
```

```
Training and tuning: LogisticRegression
Fitting 5 folds for each of 10 candidates, totalling 50 fits
```

```
C:\Users\user\AppData\Roaming\Python\Python311\site-packages\sklearn\model_select
ion\_search.py:324: UserWarning: The total space of parameters 10 is smaller than
n_iter=12. Running 10 iterations. For exhaustive searches, use GridSearchCV.
  warnings.warn(
Training and tuning: SVM
Fitting 5 folds for each of 12 candidates, totalling 60 fits

Training and tuning: RandomForest
Fitting 5 folds for each of 12 candidates, totalling 60 fits

Training and tuning: XGBoost
Fitting 5 folds for each of 12 candidates, totalling 60 fits
```

Out[ ]:

	Model	BestCV_F1	Threshold	Accuracy	Precision	Recall	F1	RC
3	XGBoost	0.631359	0.58	0.780696	0.573696	0.676471	0.620859	0.
1	SVM	0.620997	0.54	0.761533	0.537402	0.729947	0.619048	0.
0	LogisticRegression	0.633631	0.62	0.770050	0.552966	0.697861	0.617021	0.
2	RandomForest	0.634376	0.58	0.783534	0.582734	0.649733	0.614412	0.

In [ ]:

```
print(results_df)
```

```
                Model  BestCV_F1  Threshold  Accuracy  Precision    Recall  \
3             XGBoost   0.631359       0.58  0.780696   0.573696  0.676471
1                 SVM   0.620997       0.54  0.761533   0.537402  0.729947
0  LogisticRegression   0.633631       0.62  0.770050   0.552966  0.697861
2        RandomForest   0.634376       0.58  0.783534   0.582734  0.649733

         F1   ROC_AUC                                        BestParams
3  0.620859  0.836764  {'model__subsample': 0.9, 'model__n_estimators...
1  0.619048  0.832775  {'model__kernel': 'rbf', 'model__gamma': 'scal...
0  0.617021  0.842137                   {'model__C': 4.6415888336127775}
2  0.614412  0.841171  {'model__n_estimators': 200, 'model__min_sampl...
```

In [ ]:

```
best_model_name = results_df.iloc[0]["Model"]
print(best_model_name)
```

```
XGBoost
```

In [ ]:

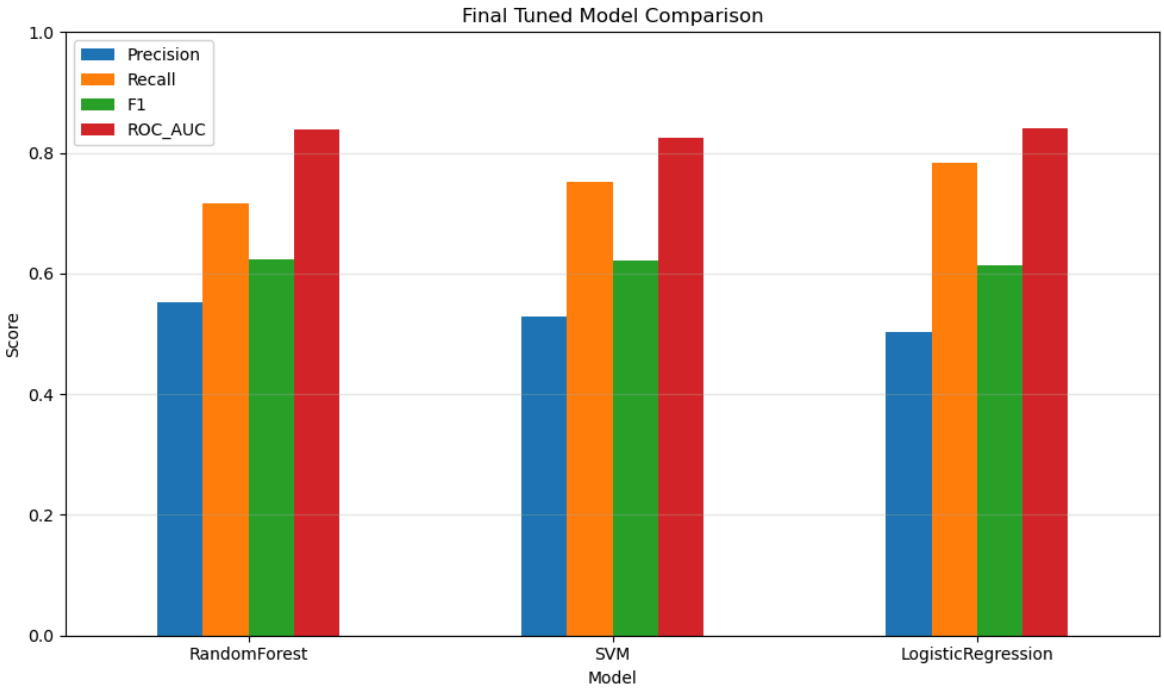
```
import joblib

joblib.dump(best_model, "best_churn_model.pkl")
joblib.dump("XGBoost Tuned", "best_model_name.pkl")
```

In [ ]:
```
results_df[["Model", "Threshold", "Precision", "Recall", "F1", "ROC_AUC", "BestC
```

Out[ ]:

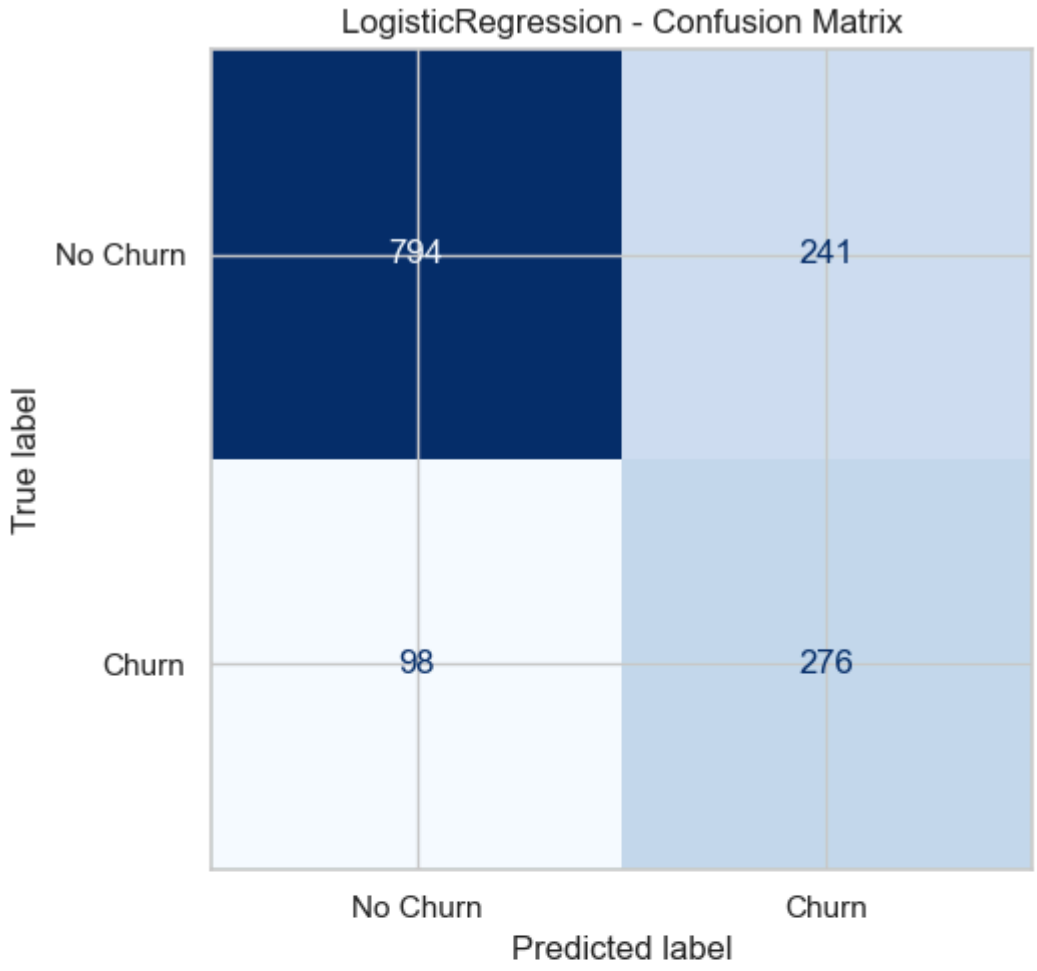
	Model	Threshold	Precision	Recall	F1	ROC_AUC	BestCV_F1
3	XGBoost	0.58	0.573696	0.676471	0.620859	0.836764	0.631359
0	LogisticRegression	0.56	0.533849	0.737968	0.619529	0.842897	0.634069
1	SVM	0.54	0.537402	0.729947	0.619048	0.832775	0.620997
2	RandomForest	0.58	0.582734	0.649733	0.614412	0.841171	0.634376

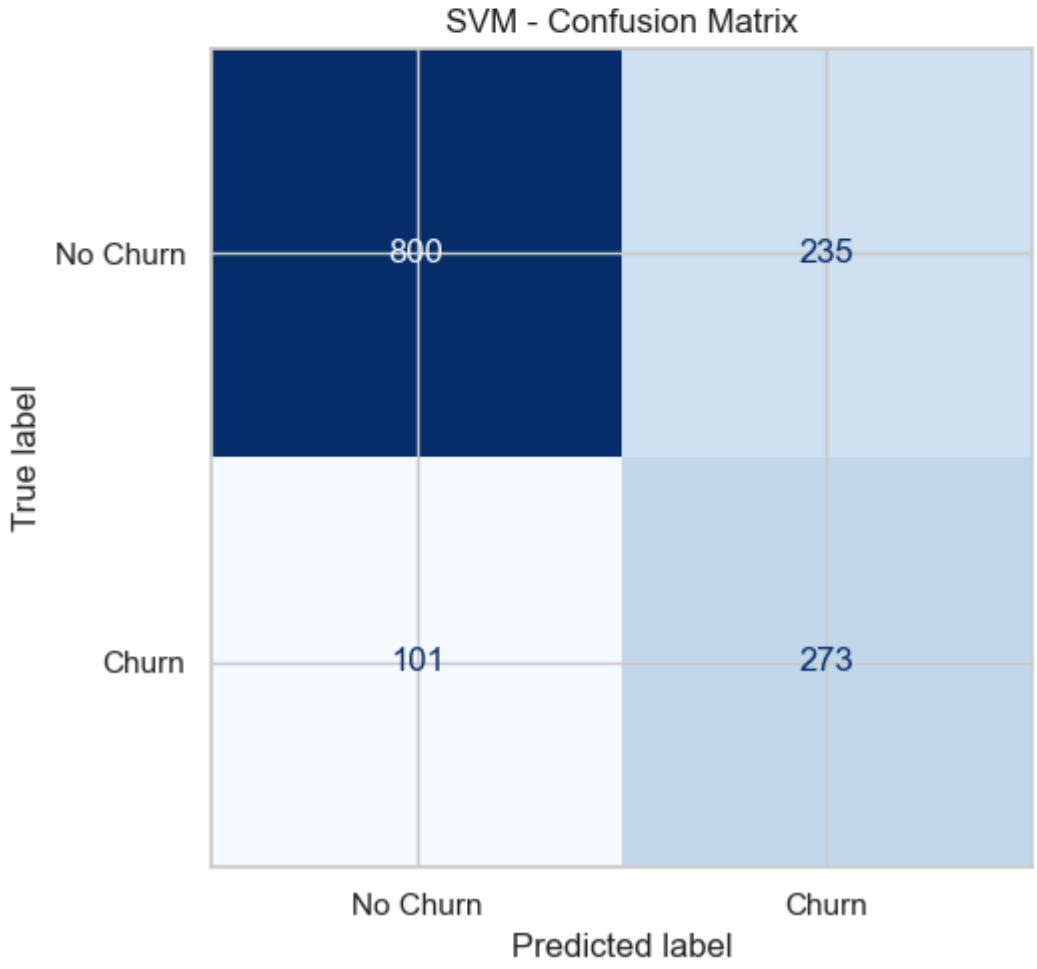
In [ ]:
```
plot_df = results_df[["Model", "Precision", "Recall", "F1", "ROC_AUC"]].set_inde

plot_df.plot(kind="bar", figsize=(10, 6))
plt.title("Final Tuned Model Comparison")
plt.ylabel("Score")
plt.xlabel("Model")
plt.xticks(rotation=0)
plt.ylim(0, 1)
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```



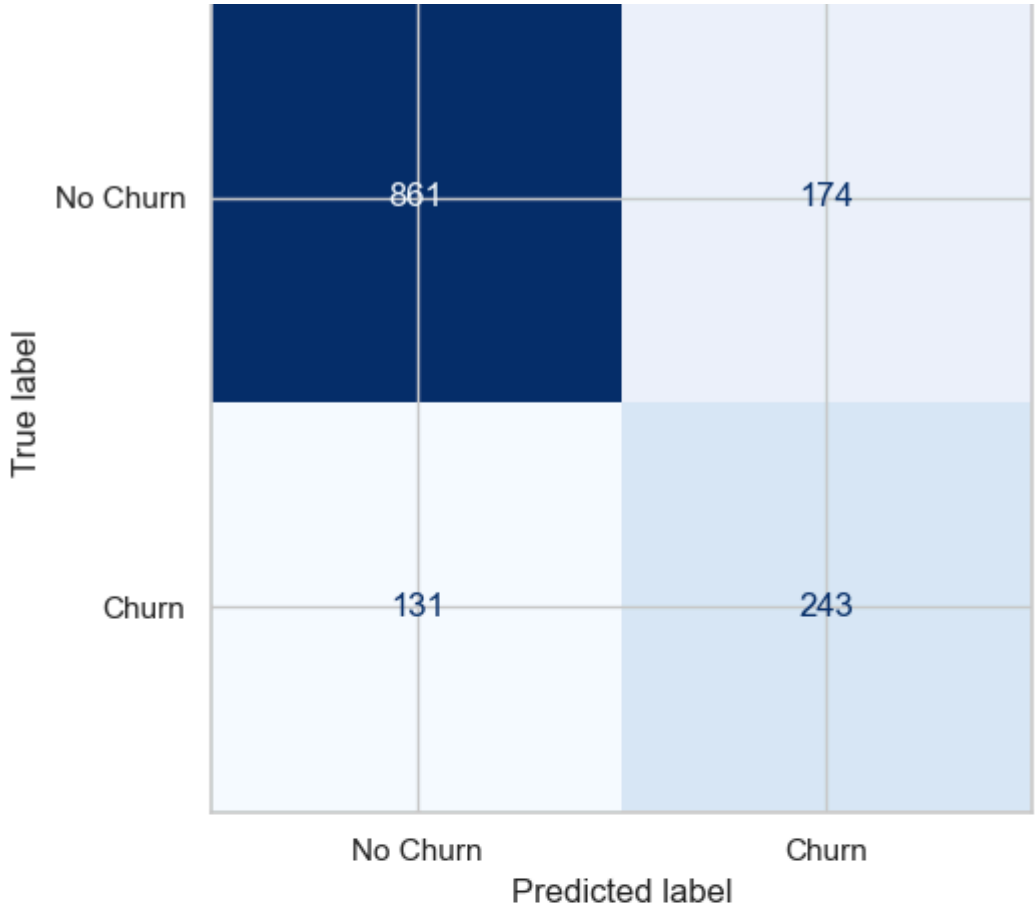
In [ ]:
```
n_models = len(artifacts)
fig, axes = plt.subplots(n_models, 1, figsize=(6, 5 * n_models))

if n_models == 1:
    axes = [axes]

for ax, (model_name, values) in zip(axes, artifacts.items()):
    disp = ConfusionMatrixDisplay(
        confusion_matrix=values["confusion_matrix"],
        display_labels=["No Churn", "Churn"],
    )
    disp.plot(ax=ax, cmap="Blues", colorbar=False)
    ax.set_title(f"{model_name} - Confusion Matrix")
```

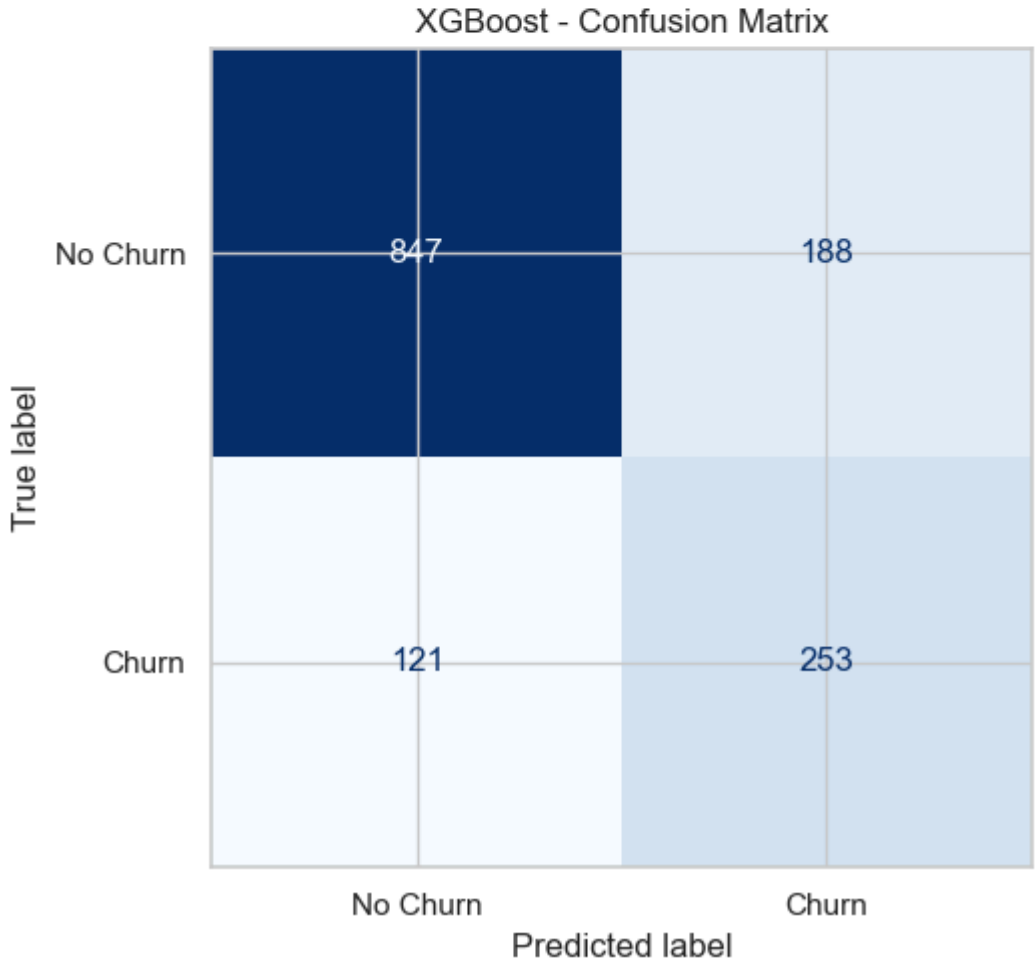
```
plt.tight_layout()
plt.show()
```


LogisticRegression - Confusion Matrix
No Churn
Churn
True label
794
241
98
276
No Churn
Churn
Predicted label


SVM - Confusion Matrix
No Churn
Churn
True label
800
235
101
273
No Churn
Churn
Predicted label

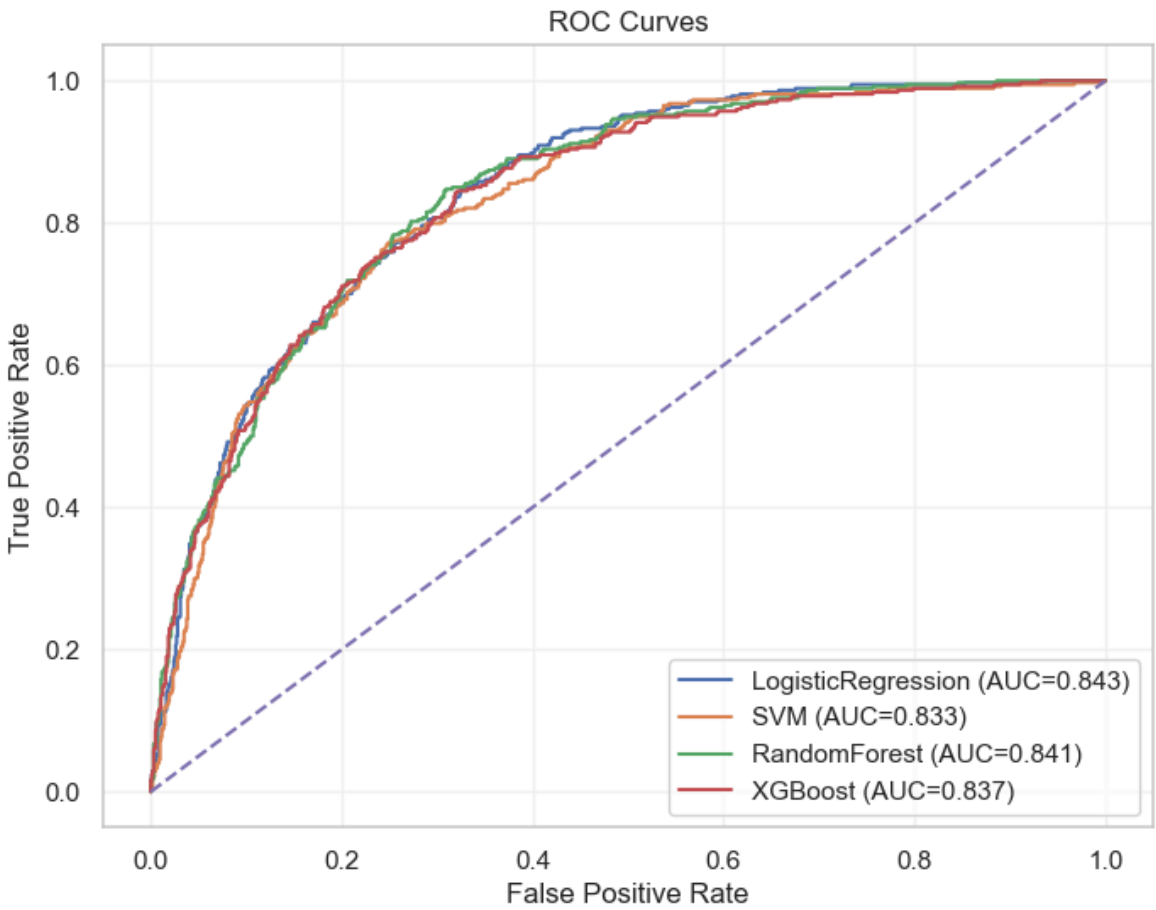
RandomForest - Confusion Matrix





In [ ]:
```
plt.figure(figsize=(8, 6))
```

```
for model_name, values in artifacts.items():
    roc_auc = auc(values["fpr"], values["tpr"])
    plt.plot(values["fpr"], values["tpr"], label=f"{model_name} (AUC={roc_auc:.3

plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves")
plt.legend()
plt.grid(alpha=0.2)
plt.show()
```



In [ ]:
```
best_model_name = results_df.iloc[0]["Model"]
best_pipeline = trained_pipelines[best_model_name]

perm = permutation_importance(
    best_pipeline,
    X_test,
    y_test,
    n_repeats=10,
    random_state=RANDOM_STATE,
    scoring="f1",
    n_jobs=-1,
)

importance_df = pd.DataFrame({
    "Feature": X_test.columns,
    "ImportanceMean": perm.importances_mean,
    "ImportanceStd": perm.importances_std,
}).sort_values(by="ImportanceMean", ascending=False)
```

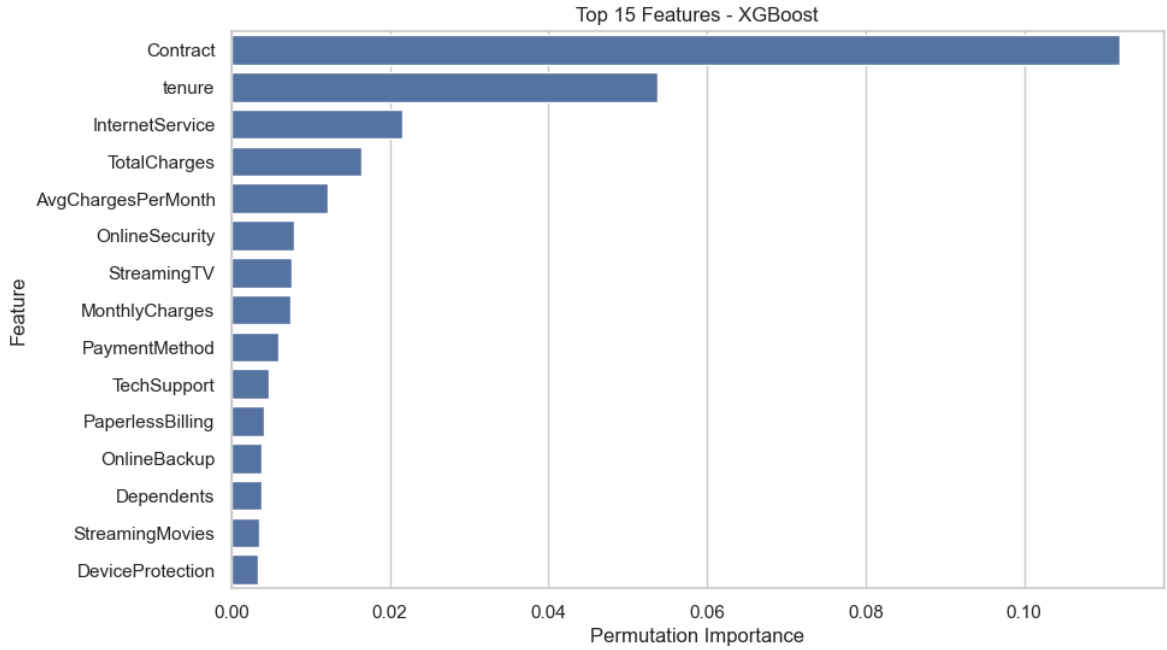
```
importance_df.head(15)
```

Out[ ]:

	Feature	ImportanceMean	ImportanceStd
14	Contract	0.112016	0.012453
4	tenure	0.053726	0.010483
7	InternetService	0.021631	0.007328
18	TotalCharges	0.016436	0.007883
19	AvgChargesPerMonth	0.012131	0.011637
8	OnlineSecurity	0.007902	0.004706
12	StreamingTV	0.007676	0.002723
17	MonthlyCharges	0.007507	0.006101
16	PaymentMethod	0.005941	0.006214
11	TechSupport	0.004775	0.006967
15	PaperlessBilling	0.004127	0.003447
9	OnlineBackup	0.003886	0.005282
3	Dependents	0.003885	0.002674
13	StreamingMovies	0.003524	0.001745
10	DeviceProtection	0.003418	0.002790

In [ ]:

```
top_features = importance_df.head(15)

plt.figure(figsize=(10, 6))
sns.barplot(data=top_features, x="ImportanceMean", y="Feature")
plt.title(f"Top 15 Features - {best_model_name}")
plt.xlabel("Permutation Importance")
plt.ylabel("Feature")
plt.show()
```



In [ ]:
```
df.shape
```

Out[ ]:
```
(7032, 20)
```

In [ ]:
```
df['Churn'].value_counts()
```

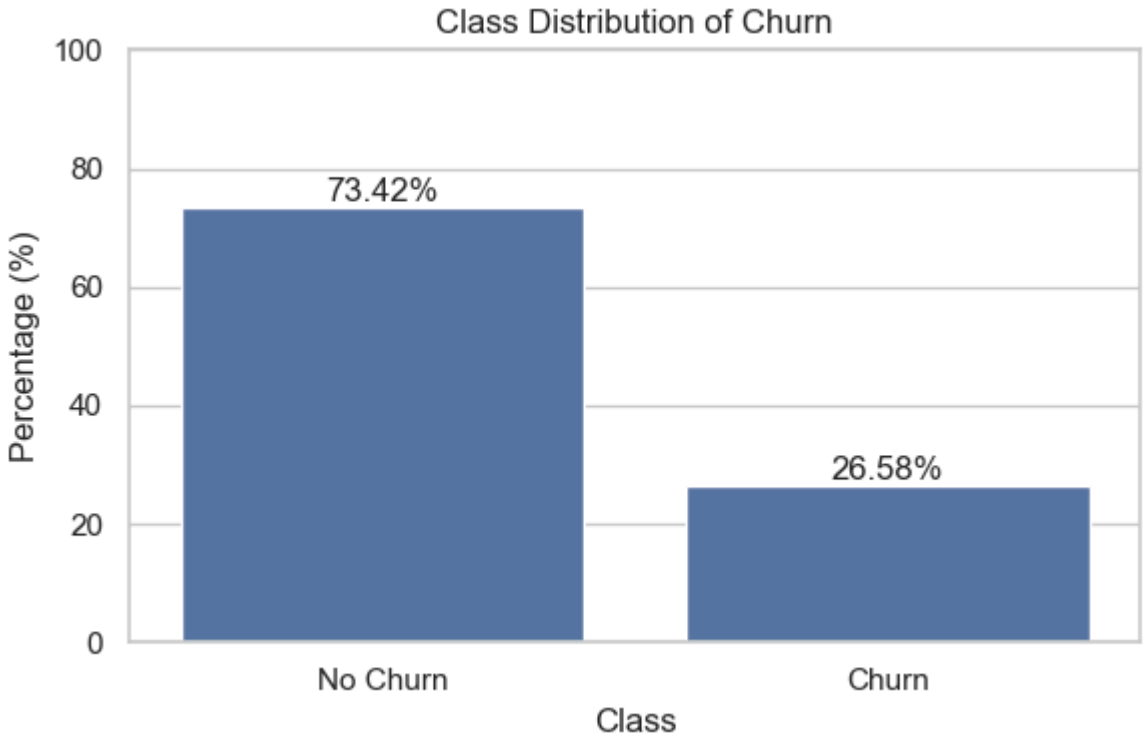
Out[ ]:
```
Churn
0    5163
1    1869
Name: count, dtype: int64
```

In [ ]:
```
from sklearn.preprocessing import OneHotEncoder

# Separate features and target
X = df.drop('Churn', axis=1)
y = df['Churn']

# Identify numeric and categorical columns
numeric_features = X.select_dtypes(include=['int64', 'float64']).columns.tolist(
categorical_features = X.select_dtypes(include=['object', 'string']).columns.tol

print("Numeric features:", numeric_features)
print("Categorical features:", categorical_features)

# Preprocessing transformers
numeric_transformer = StandardScaler()

# One-hot encode categorical variables
categorical_transformer = OneHotEncoder(drop='first', handle_unknown='ignore')

# Build preprocessing pipeline
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, numeric_features),
        ('cat', categorical_transformer, categorical_features)
    ],
    remainder='drop'
)
```

```
print("Preprocessor defined.")
```

```
Numeric features: ['SeniorCitizen', 'tenure', 'MonthlyCharges']
Categorical features: ['customerID', 'gender', 'Partner', 'Dependents', 'PhoneSer
vice', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'Dev
iceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'Pap
erlessBilling', 'PaymentMethod', 'TotalCharges']
Preprocessor defined.
```

In [ ]:
```
churn_dist = df['Churn'].value_counts(normalize=True) * 100

plt.figure(figsize=(6,4))
sns.barplot(x=churn_dist.index, y=churn_dist.values)
plt.xticks([0, 1], ['No Churn', 'Churn'])
plt.ylabel('Percentage (%)')
plt.xlabel('Class')
plt.title('Class Distribution of Churn')
plt.ylim(0, 100)

for i, v in enumerate(churn_dist.values):
    plt.text(i, v + 1, f"{v:.2f}%", ha='center')

plt.tight_layout()
plt.show()
```



In [ ]:
```
# Train-test split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Train shape:", X_train.shape, "Test shape:", X_test.shape)
```

```
Train shape: (5625, 19) Test shape: (1407, 19)
```

In [ ]:
```
print("y_train distribution:")
print(y_train.value_counts(normalize=True))

print("\ny_test distribution:")
print(y_test.value_counts(normalize=True))
```

```
y_train distribution:
Churn
0    0.734222
1    0.265778
Name: proportion, dtype: float64

y_test distribution:
Churn
0    0.734186
1    0.265814
Name: proportion, dtype: float64
```

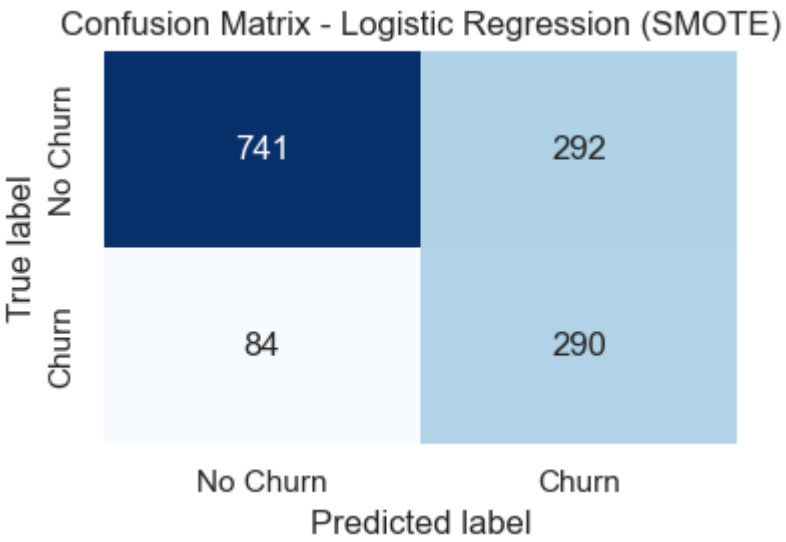
In [ ]:
```
def evaluate_model(name, model, X_test, y_test):
    y_pred = model.predict(X_test)
    y_proba = None

    if hasattr(model, "predict_proba"):
        y_proba = model.predict_proba(X_test)[:, 1]
    elif hasattr(model, "decision_function"):
        scores = model.decision_function(X_test)
        y_proba = (scores - scores.min()) / (scores.max() - scores.min()) if sco

    cm = confusion_matrix(y_test, y_pred)
    acc = accuracy_score(y_test, y_pred)
    precision, recall, f1, _ = precision_recall_fscore_support(
        y_test, y_pred, average='binary'
    )

    print(f"\n=== {name} ===")
    print("Confusion matrix:\n", cm)
    print("Accuracy:", acc)
    print("Classification report:\n", classification_report(y_test, y_pred))

    plt.figure(figsize=(4, 3))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False,
                xticklabels=['No Churn', 'Churn'],
                yticklabels=['No Churn', 'Churn'])
    plt.xlabel('Predicted label')
    plt.ylabel('True label')
    plt.title(f'Confusion Matrix - {name}')
    plt.tight_layout()
    plt.show()

    auc = None
    if y_proba is not None:
        auc = roc_auc_score(y_test, y_proba)
        print("ROC-AUC:", auc)

        fpr, tpr, _ = roc_curve(y_test, y_proba)
        plt.figure(figsize=(6, 4))
        plt.plot(fpr, tpr, label=f"{name} (AUC = {auc:.3f})")
        plt.plot([0, 1], [0, 1], 'k--')
        plt.xlabel("False Positive Rate")
        plt.ylabel("True Positive Rate")
```

```
        plt.title(f"ROC Curve - {name}")
        plt.legend(loc="lower right")
        plt.show()

    metrics = {
        'model': name,
        'accuracy': acc,
        'precision': precision,
        'recall': recall,
        'f1': f1,
        'roc_auc': auc
    }

    return metrics
```

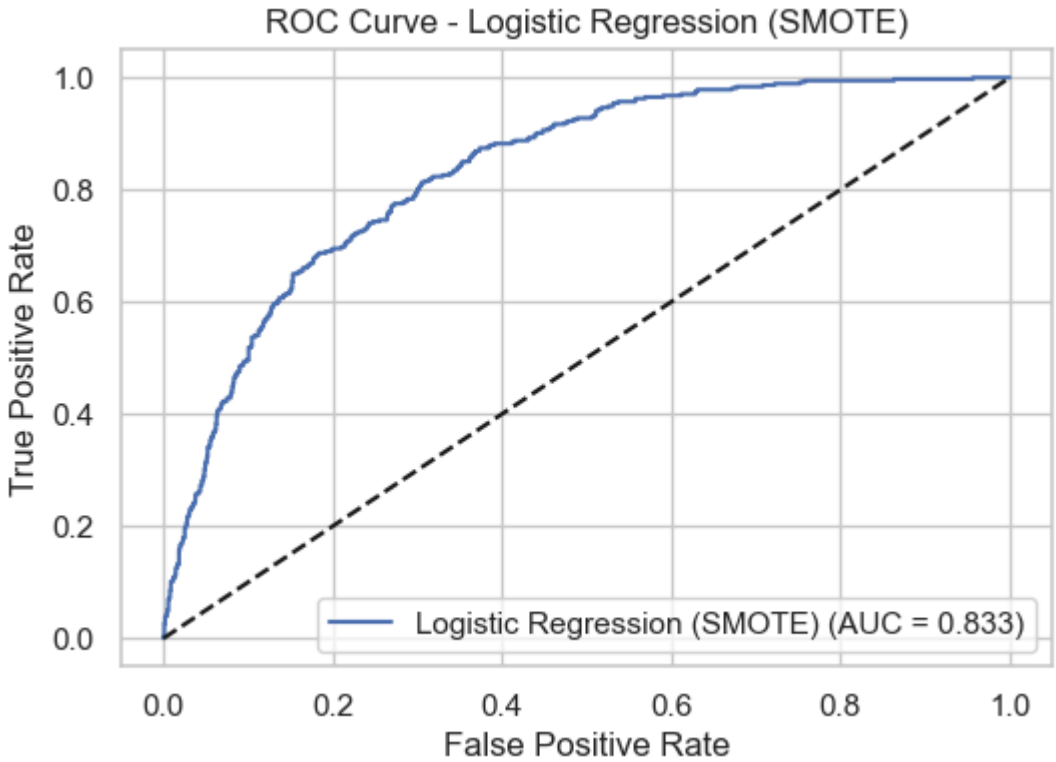
In [ ]:

```
results = []

log_reg = ImbPipeline(steps=[
    ('preprocess', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', LogisticRegression(max_iter=1000, random_state=42))
])

log_reg.fit(X_train, y_train)

metrics_log_reg = evaluate_model("Logistic Regression (SMOTE)", log_reg, X_test,
results.append(metrics_log_reg)
```

```
=== Logistic Regression (SMOTE) ===
Confusion matrix:
 [[741 292]
 [ 84 290]]
Accuracy: 0.7327647476901208
Classification report:
              precision    recall  f1-score   support

           0       0.90      0.72      0.80      1033
           1       0.50      0.78      0.61       374

    accuracy                           0.73      1407
   macro avg       0.70      0.75      0.70      1407
weighted avg       0.79      0.73      0.75      1407
```

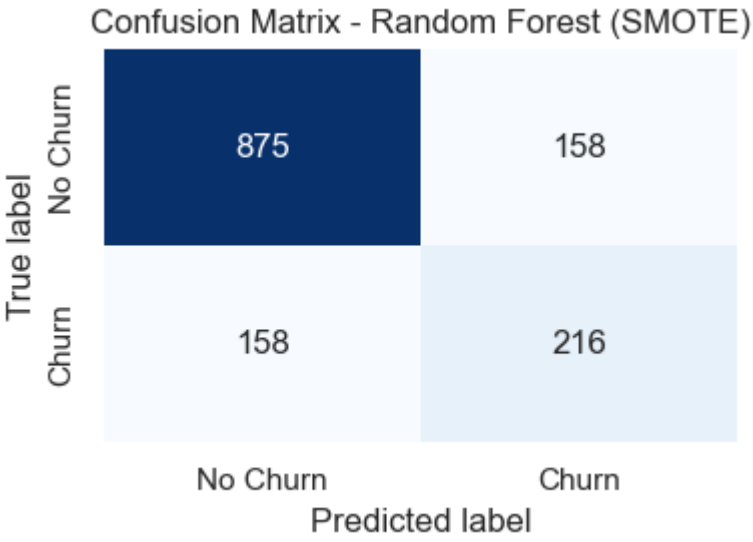


```
ROC-AUC: 0.8332060713046989
```

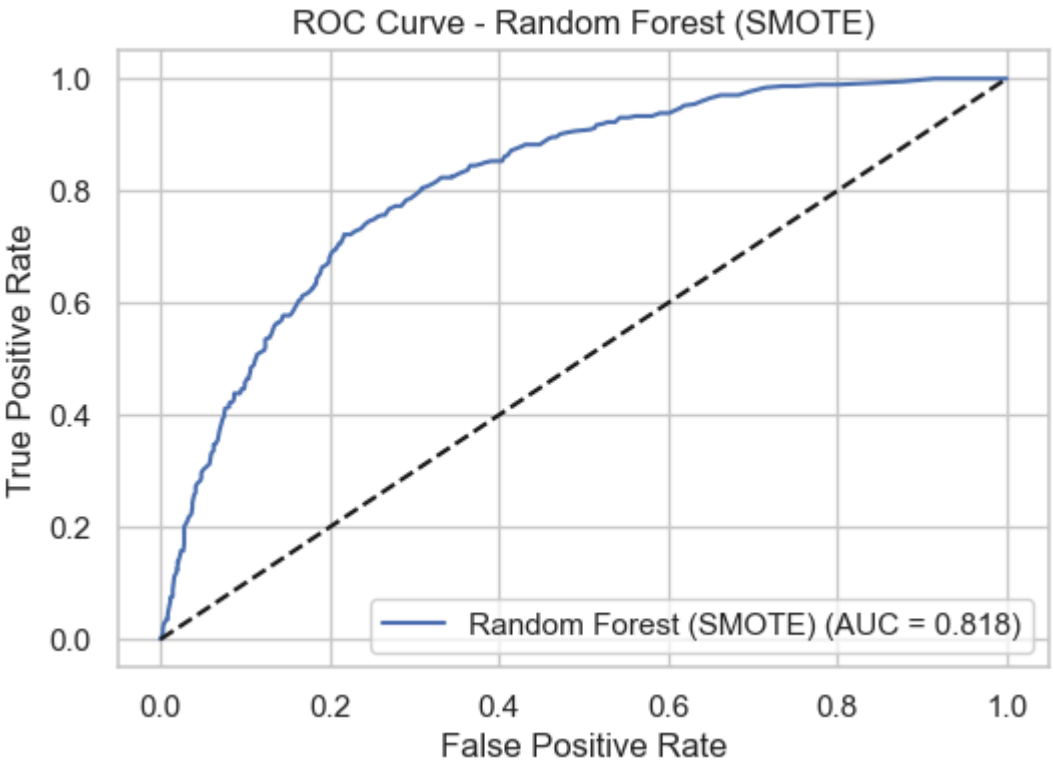


In [ ]:
```
# 2. Random Forest with SMOTE

rf_clf = ImbPipeline(steps=[
    ('preprocess', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', RandomForestClassifier(n_estimators=200, random_state=42))
])

rf_clf.fit(X_train, y_train)

metrics_rf = evaluate_model("Random Forest (SMOTE)", rf_clf, X_test, y_test)
results.append(metrics_rf)
```

```
=== Random Forest (SMOTE) ===
Confusion matrix:
 [[875 158]
 [158 216]]
Accuracy: 0.775408670931059
Classification report:
              precision    recall  f1-score   support

           0       0.85      0.85      0.85      1033
           1       0.58      0.58      0.58       374

    accuracy                           0.78      1407
   macro avg       0.71      0.71      0.71      1407
weighted avg       0.78      0.78      0.78      1407
```




```
ROC-AUC: 0.8175877849159552
```



In [ ]:
```
from sklearn.model_selection import RandomizedSearchCV
```

In [ ]:
```
# Random Forest tuning

rf_tune_pipeline = ImbPipeline(steps=[
    ('preprocess', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', RandomForestClassifier(random_state=42))
])

rf_param_dist = {
    'clf__n_estimators': [100, 200, 300, 500],
    'clf__max_depth': [None, 5, 10, 20, 30],
    'clf__min_samples_split': [2, 5, 10],
    'clf__min_samples_leaf': [1, 2, 4],
    'clf__max_features': ['sqrt', 'log2', None]
}
```

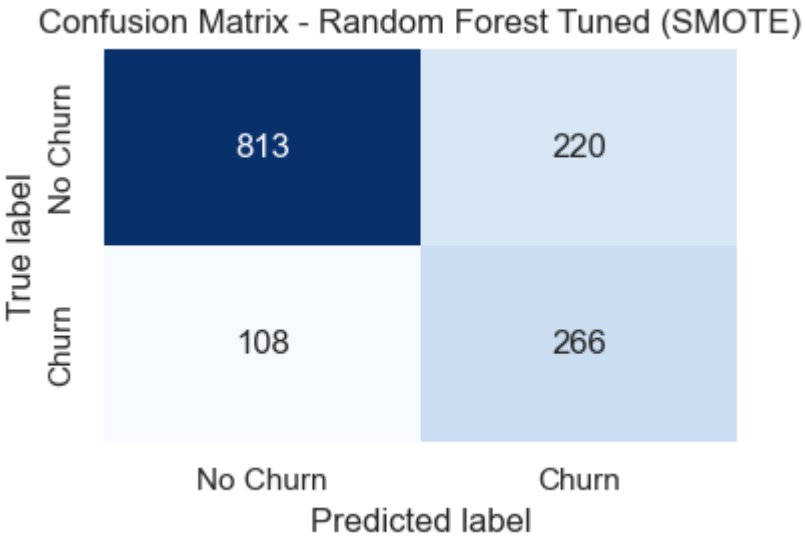
```
rf_random_search = RandomizedSearchCV(
    estimator=rf_tune_pipeline,
    param_distributions=rf_param_dist,
    n_iter=15,
    scoring='roc_auc',
    cv=5,
    verbose=2,
    random_state=42,
    n_jobs=-1
)

rf_random_search.fit(X_train, y_train)

print("Best Random Forest parameters:")
print(rf_random_search.best_params_)

print("\nBest Random Forest ROC-AUC:")
print(rf_random_search.best_score_)
```

```
Fitting 5 folds for each of 15 candidates, totalling 75 fits
Best Random Forest parameters:
{'clf__n_estimators': 100, 'clf__min_samples_split': 10, 'clf__min_samples_leaf':
2, 'clf__max_features': 'log2', 'clf__max_depth': 10}

Best Random Forest ROC-AUC:
0.8434316972636797
```

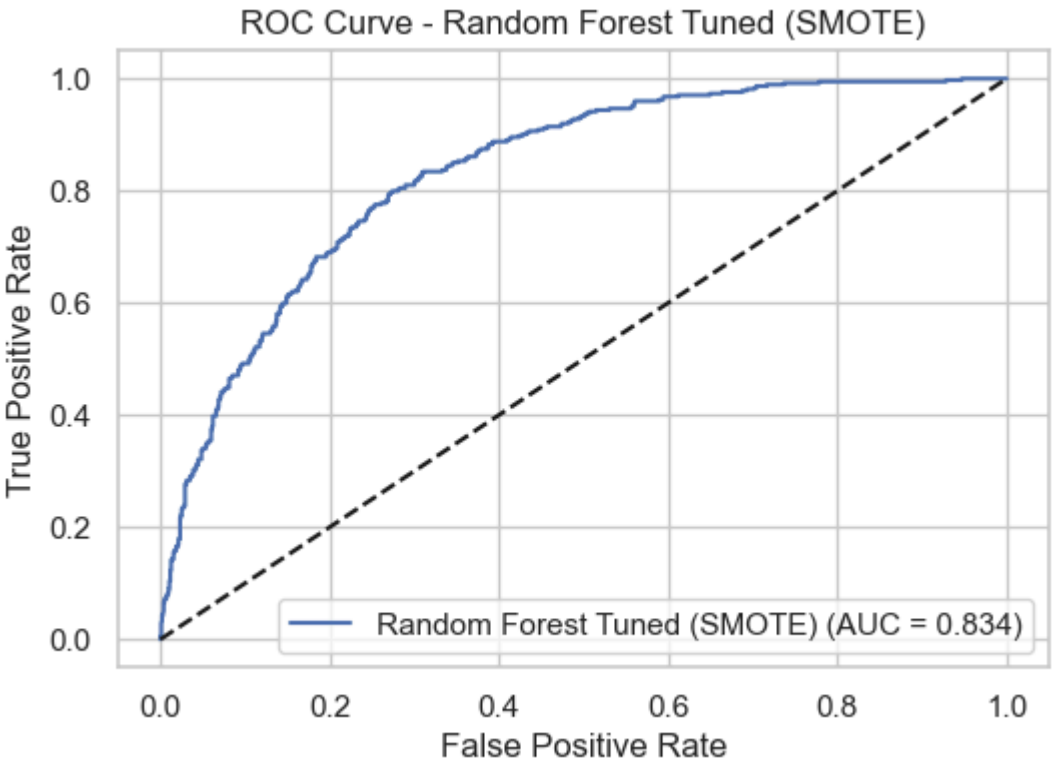
In [ ]:

```
best_rf_model = rf_random_search.best_estimator_

metrics_rf_tuned = evaluate_model(
    "Random Forest Tuned (SMOTE)",
    best_rf_model,
    X_test,
    y_test
)

results.append(metrics_rf_tuned)
```

```
=== Random Forest Tuned (SMOTE) ===
Confusion matrix:
 [[813 220]
 [108 266]]
Accuracy: 0.7668798862828714
Classification report:
               precision    recall  f1-score   support

           0       0.88      0.79      0.83      1033
           1       0.55      0.71      0.62       374

    accuracy                           0.77      1407
   macro avg       0.72      0.75      0.73      1407
weighted avg       0.79      0.77      0.78      1407
```



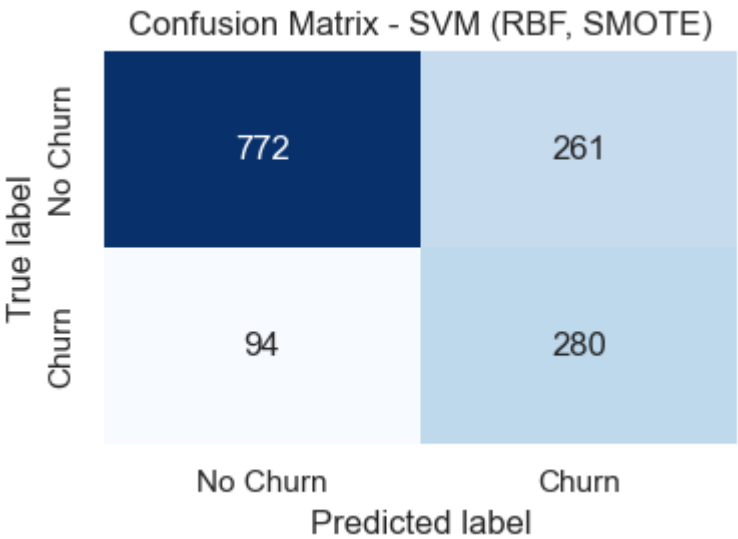

```
ROC-AUC: 0.8344251983993456
```



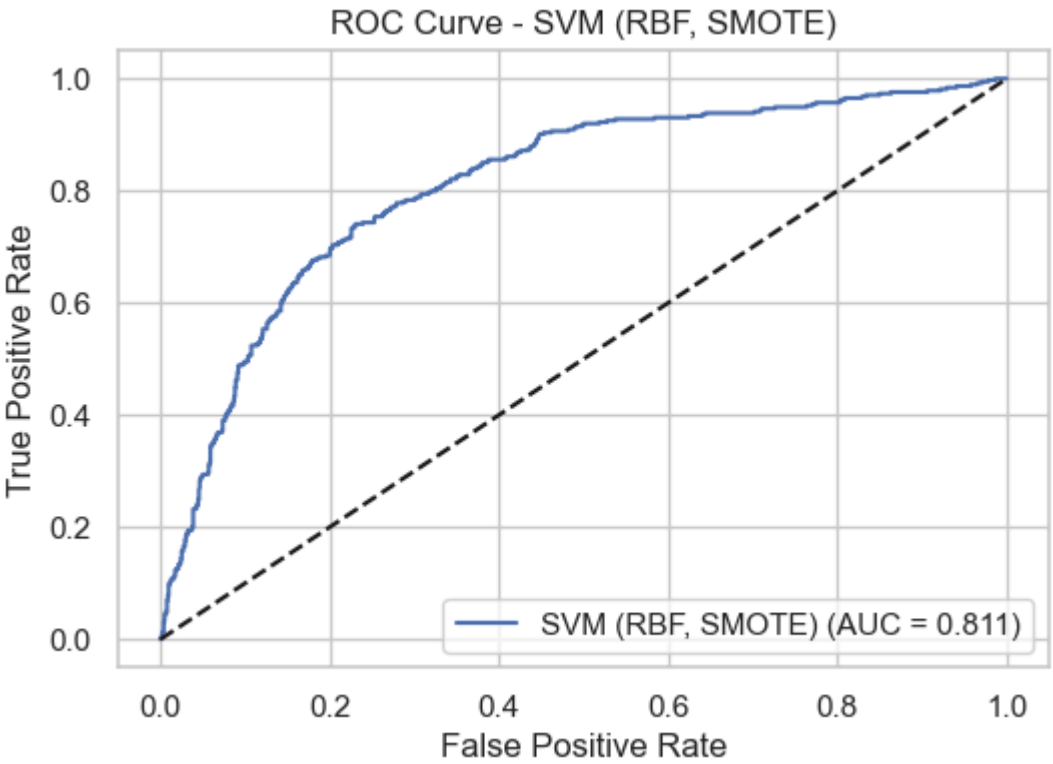
In [ ]:

```python
# 3. Support Vector Machine (SVM) with SMOTE

svm_clf = ImbPipeline(steps=[
    ('preprocess', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', SVC(kernel='rbf', probability=True, random_state=42))
])

svm_clf.fit(X_train, y_train)

metrics_svm = evaluate_model("SVM (RBF, SMOTE)", svm_clf, X_test, y_test)
results.append(metrics_svm)
```

```
=== SVM (RBF, SMOTE) ===
Confusion matrix:
 [[772 261]
 [ 94 280]]
Accuracy: 0.7476901208244492
Classification report:
              precision    recall  f1-score   support

           0       0.89      0.75      0.81      1033
           1       0.52      0.75      0.61       374

    accuracy                           0.75      1407
   macro avg       0.70      0.75      0.71      1407
weighted avg       0.79      0.75      0.76      1407
```



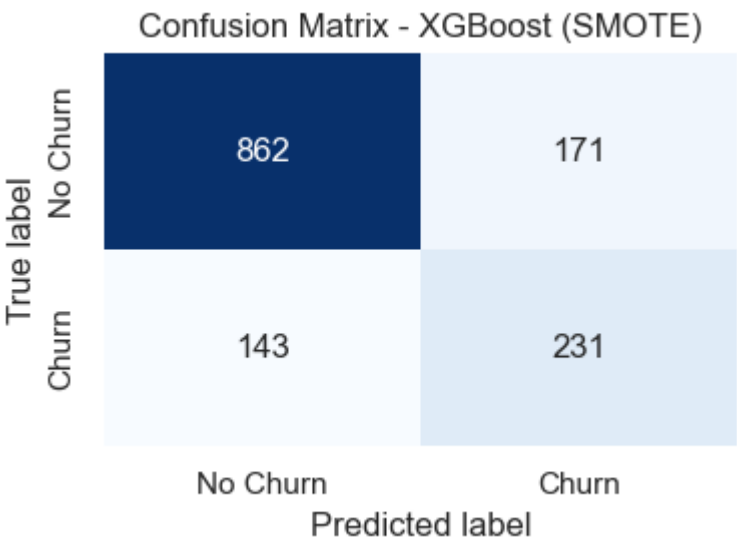

```
ROC-AUC: 0.810738930791889
```



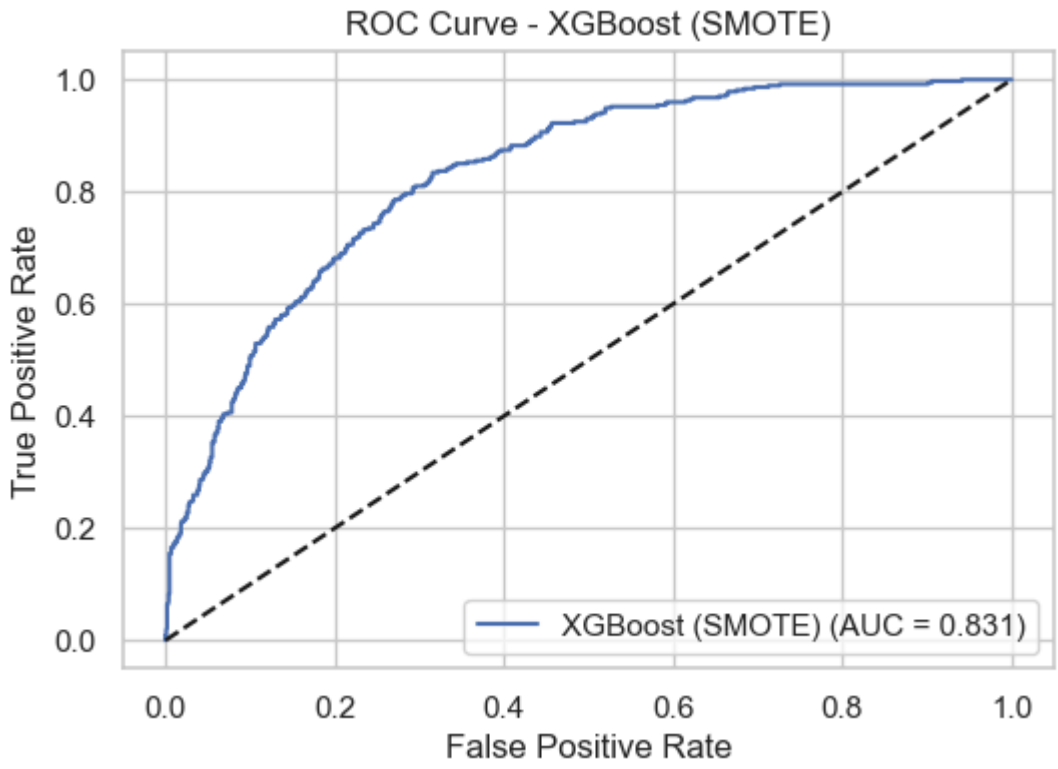
In [ ]:
```
# 4. XGBoost (if available) with SMOTE

if xgb_available:
```

```
    xgb_clf = ImbPipeline(steps=[
        ('preprocess', preprocessor),
        ('smote', SMOTE(random_state=42)),
        ('clf', XGBClassifier(
            n_estimators=300,
            learning_rate=0.05,
            max_depth=4,
            subsample=0.8,
            colsample_bytree=0.8,
            objective='binary:logistic',
            eval_metric='logloss',
            random_state=42
        ))
    ])

    xgb_clf.fit(X_train, y_train)

    metrics_xgb = evaluate_model("XGBoost (SMOTE)", xgb_clf, X_test, y_test)
    results.append(metrics_xgb)
else:
    print("XGBoost is not installed. Install it with 'pip install xgboost' to ru
```

```
=== XGBoost (SMOTE) ===
Confusion matrix:
 [[862 171]
 [143 231]]
Accuracy: 0.7768301350390903
Classification report:
              precision    recall  f1-score   support

           0       0.86      0.83      0.85      1033
           1       0.57      0.62      0.60       374

    accuracy                           0.78      1407
   macro avg       0.72      0.73      0.72      1407
weighted avg       0.78      0.78      0.78      1407
```




```
ROC-AUC: 0.8310460679915722
```



In [ ]:
```
# XGBoost tuning

if xgb_available:
    xgb_tune_pipeline = ImbPipeline(steps=[
        ('preprocess', preprocessor),
        ('smote', SMOTE(random_state=42)),
        ('clf', XGBClassifier(
            objective='binary:logistic',
            eval_metric='logloss',
            random_state=42
        ))
    ])

    xgb_param_dist = {
        'clf__n_estimators': [100, 200, 300, 500],
        'clf__learning_rate': [0.01, 0.05, 0.1, 0.2],
        'clf__max_depth': [3, 4, 5, 6],
        'clf__subsample': [0.7, 0.8, 0.9, 1.0],
        'clf__colsample_bytree': [0.7, 0.8, 0.9, 1.0],
        'clf__gamma': [0, 0.1, 0.3, 0.5]
    }

    xgb_random_search = RandomizedSearchCV(
        estimator=xgb_tune_pipeline,
        param_distributions=xgb_param_dist,
        n_iter=15,
        scoring='roc_auc',
        cv=5,
        verbose=2,
        random_state=42,
        n_jobs=-1
    )

    xgb_random_search.fit(X_train, y_train)
```

```
    print("Best XGBoost parameters:")
    print(xgb_random_search.best_params_)

    print("\nBest XGBoost ROC-AUC:")
    print(xgb_random_search.best_score_)
else:
    print("XGBoost is not available.")
```

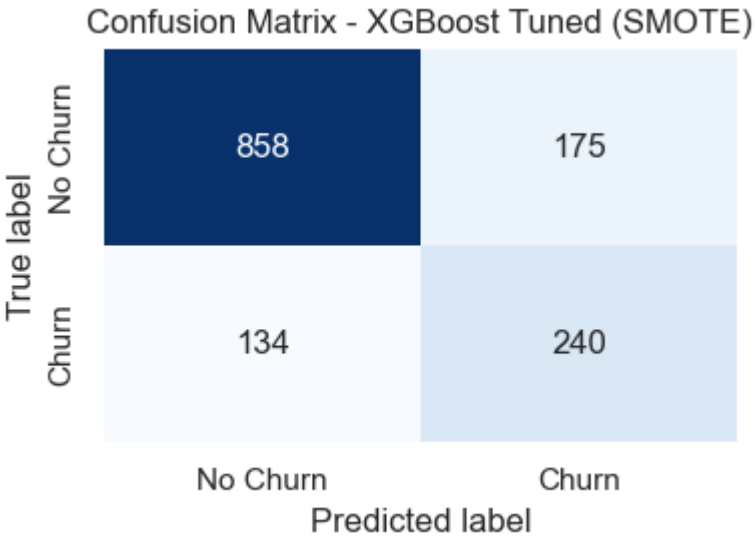
```
Fitting 5 folds for each of 15 candidates, totalling 75 fits
```

```
C:\Users\user\AppData\Roaming\Python\Python311\site-packages\sklearn\model_select
ion\_search.py:1137: UserWarning: One or more of the test scores are non-finite:
[nan nan nan nan nan nan nan nan nan nan nan nan nan nan nan]
  warnings.warn(
Best XGBoost parameters:
{'clf__subsample': 0.9, 'clf__n_estimators': 200, 'clf__max_depth': 5, 'clf__lear
ning_rate': 0.05, 'clf__gamma': 0, 'clf__colsample_bytree': 1.0}

Best XGBoost ROC-AUC:
nan
```

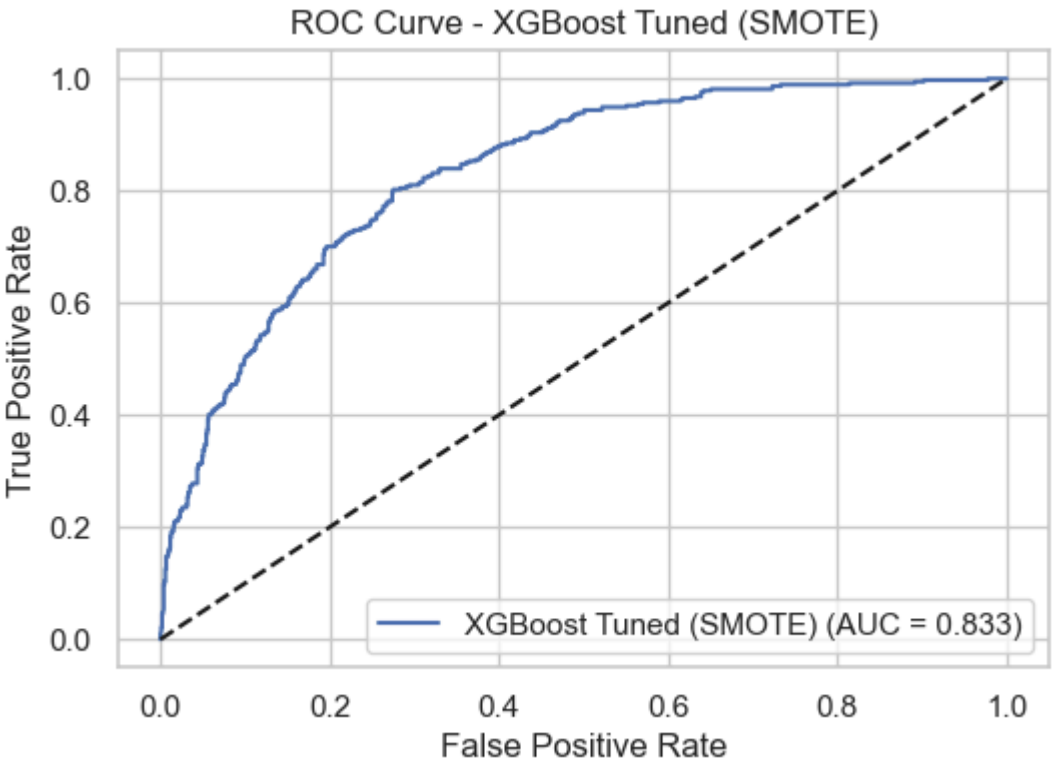
In [ ]:

```
if xgb_available:
    best_xgb_model = xgb_random_search.best_estimator_

    metrics_xgb_tuned = evaluate_model(
        "XGBoost Tuned (SMOTE)",
        best_xgb_model,
        X_test,
        y_test
    )

    results.append(metrics_xgb_tuned)
```

```
=== XGBoost Tuned (SMOTE) ===
Confusion matrix:
 [[858 175]
 [134 240]]
Accuracy: 0.7803837953091685
Classification report:
              precision    recall  f1-score   support

           0       0.86      0.83      0.85      1033
           1       0.58      0.64      0.61       374

    accuracy                           0.78      1407
   macro avg       0.72      0.74      0.73      1407
weighted avg       0.79      0.78      0.78      1407
```




```
ROC-AUC: 0.833096065144354
```



In [ ]:
```
results_df = pd.DataFrame(results)
results_df.sort_values(by='roc_auc', ascending=False)
```

Out[ ]:

	model	accuracy	precision	recall	f1	roc_auc
6	Random Forest Tuned (SMOTE)	0.766880	0.547325	0.711230	0.618605	0.834425
5	Random Forest Tuned (SMOTE)	0.766880	0.547325	0.711230	0.618605	0.834425
0	Logistic Regression (SMOTE)	0.732765	0.498282	0.775401	0.606695	0.833206
7	XGBoost Tuned (SMOTE)	0.780384	0.578313	0.641711	0.608365	0.833096
4	XGBoost (SMOTE)	0.776830	0.574627	0.617647	0.595361	0.831046
1	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
2	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
3	SVM (RBF, SMOTE)	0.747690	0.517560	0.748663	0.612022	0.810739

In [ ]:

```
# Model comparison diagrams (metrics across models)

results_df = pd.DataFrame(results)
results_df
```

Out[ ]:

	model	accuracy	precision	recall	f1	roc_auc
0	Logistic Regression (SMOTE)	0.732765	0.498282	0.775401	0.606695	0.833206
1	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
2	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
3	SVM (RBF, SMOTE)	0.747690	0.517560	0.748663	0.612022	0.810739
4	XGBoost (SMOTE)	0.776830	0.574627	0.617647	0.595361	0.831046
5	Random Forest Tuned (SMOTE)	0.766880	0.547325	0.711230	0.618605	0.834425
6	Random Forest Tuned (SMOTE)	0.766880	0.547325	0.711230	0.618605	0.834425
7	XGBoost Tuned (SMOTE)	0.780384	0.578313	0.641711	0.608365	0.833096

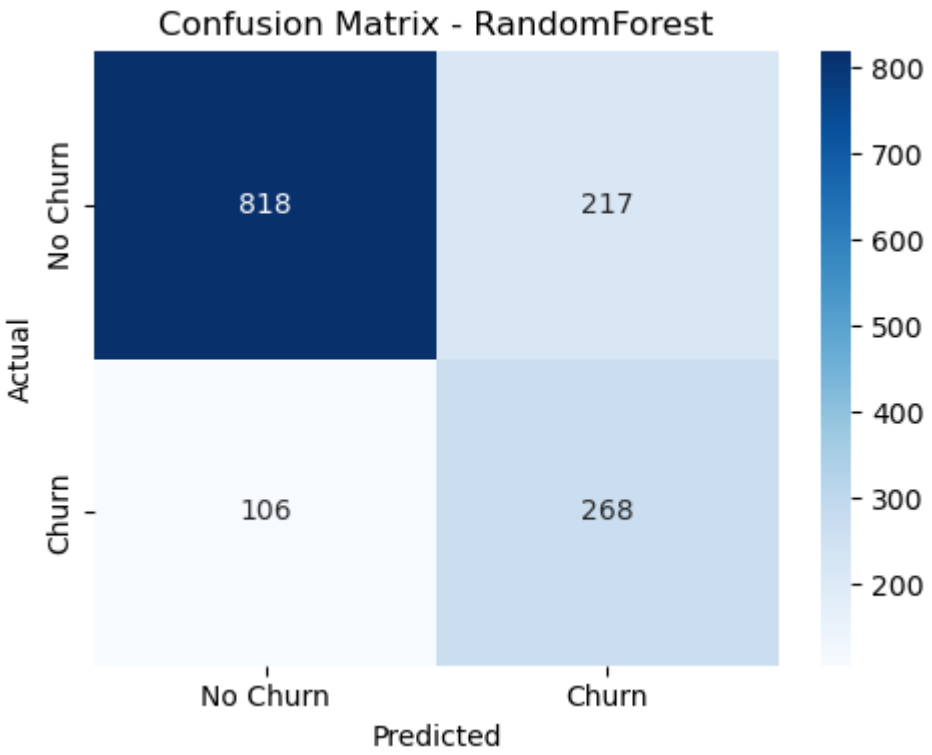
In [ ]:

```
results_df.sort_values(by="roc_auc", ascending=False)
```

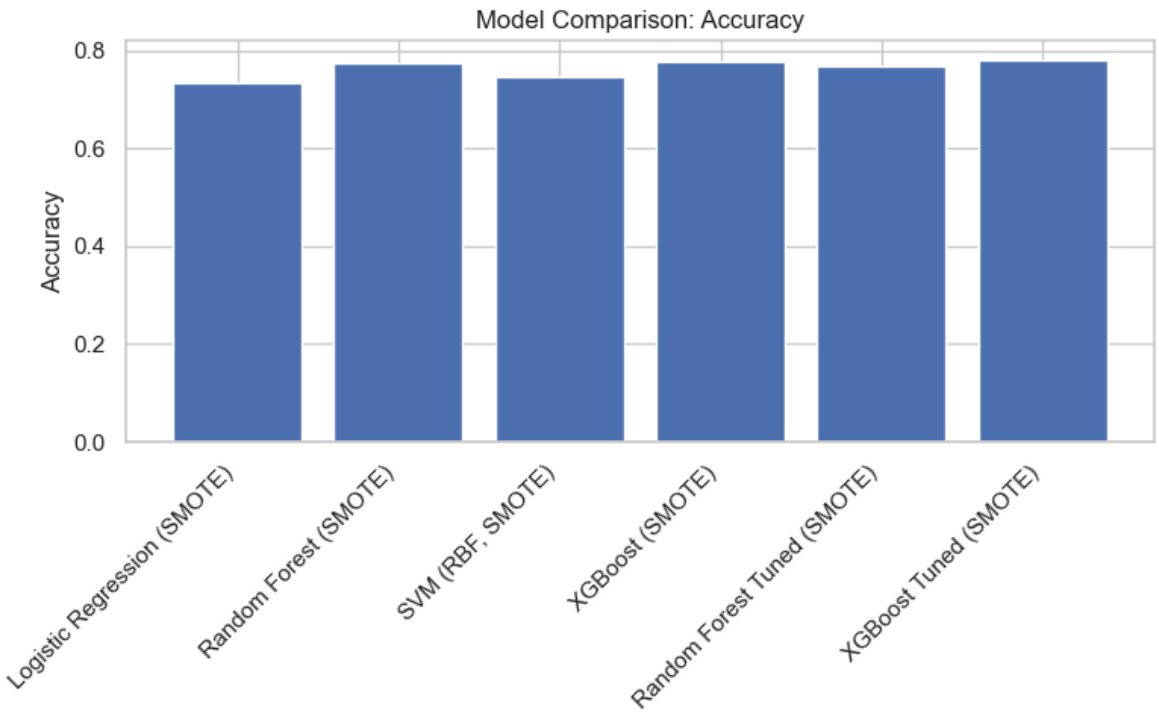
Out[ ]:

	model	accuracy	precision	recall	f1	roc_auc
6	Random Forest Tuned (SMOTE)	0.766880	0.547325	0.711230	0.618605	0.834425
5	Random Forest Tuned (SMOTE)	0.766880	0.547325	0.711230	0.618605	0.834425
0	Logistic Regression (SMOTE)	0.732765	0.498282	0.775401	0.606695	0.833206
7	XGBoost Tuned (SMOTE)	0.780384	0.578313	0.641711	0.608365	0.833096
4	XGBoost (SMOTE)	0.776830	0.574627	0.617647	0.595361	0.831046
1	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
2	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
3	SVM (RBF, SMOTE)	0.747690	0.517560	0.748663	0.612022	0.810739

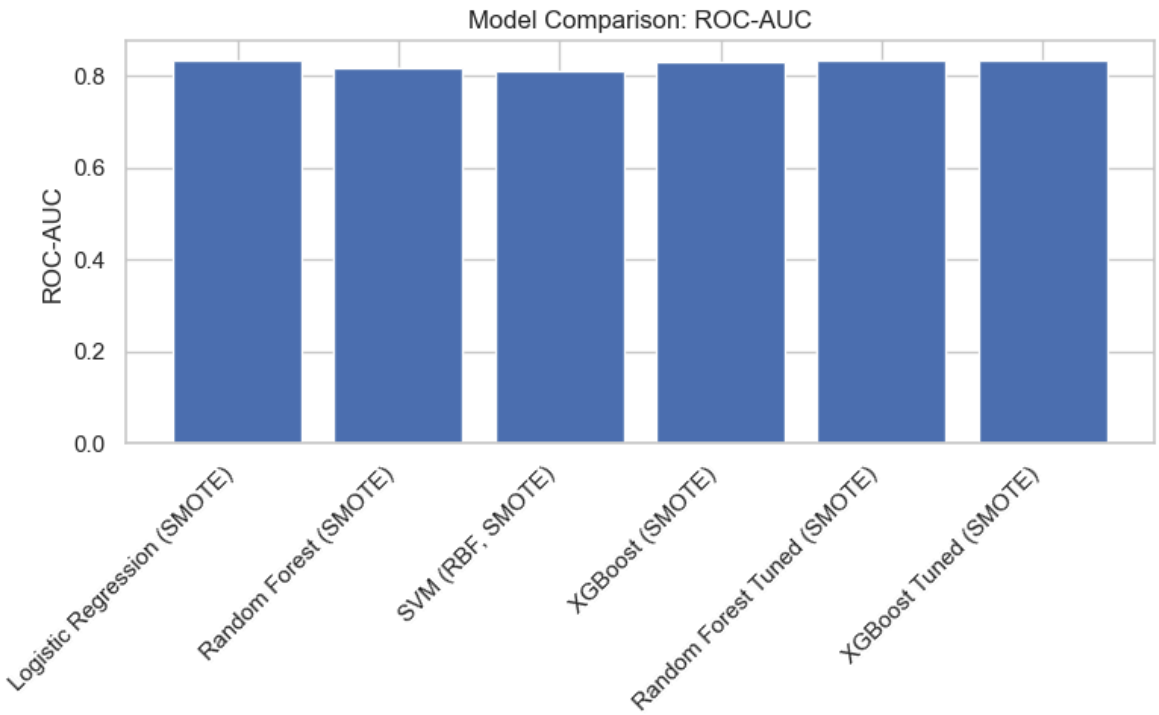
In [ ]:
```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

best_model_name = results_df.iloc[0]["Model"]
best_model = best_estimators[best_model_name]

y_pred_best = best_model.predict(X_test)
cm = confusion_matrix(y_test, y_pred_best)

plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["No Churn", "Churn"],
            yticklabels=["No Churn", "Churn"])
plt.title(f"Confusion Matrix - {best_model_name}")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()
```



In [ ]:
```
plt.figure(figsize=(8, 5))
plt.bar(results_df['model'], results_df['accuracy'])
plt.xticks(rotation=45, ha='right')
plt.ylabel('Accuracy')
plt.title('Model Comparison: Accuracy')
plt.tight_layout()
plt.show()
```

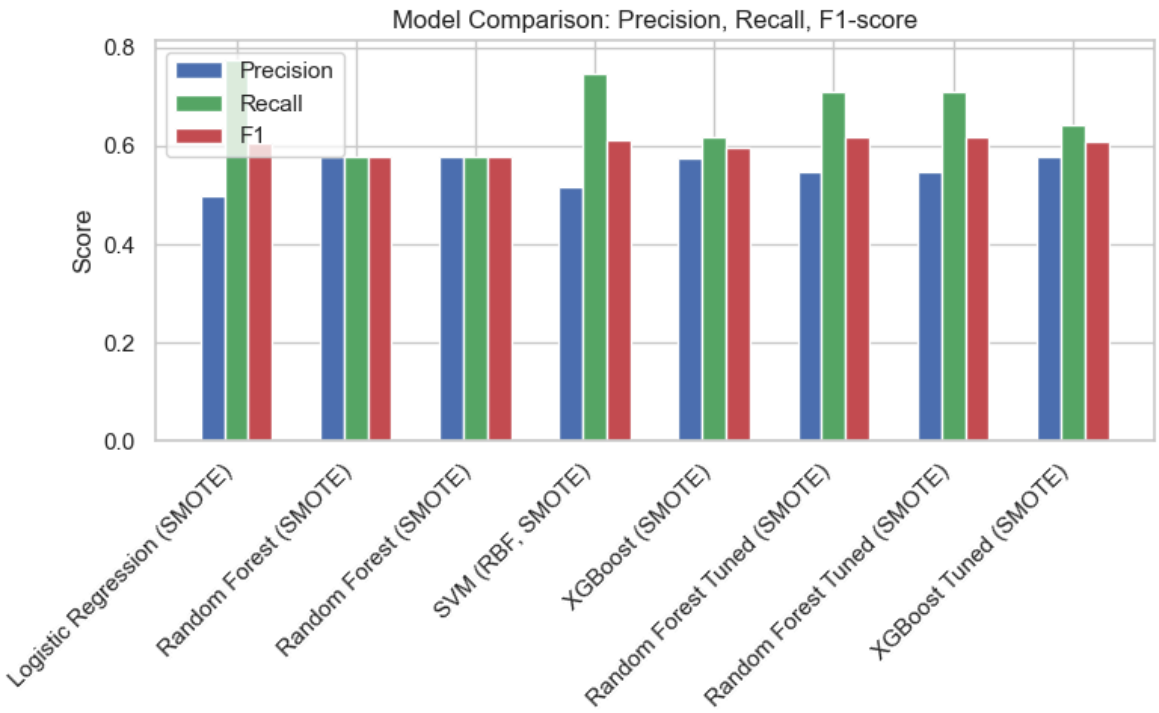


In [ ]:
```
roc_df = results_df.dropna(subset=['roc_auc'])

plt.figure(figsize=(8, 5))
plt.bar(roc_df['model'], roc_df['roc_auc'])
plt.xticks(rotation=45, ha='right')
plt.ylabel('ROC-AUC')
plt.title('Model Comparison: ROC-AUC')
plt.tight_layout()
plt.show()
```



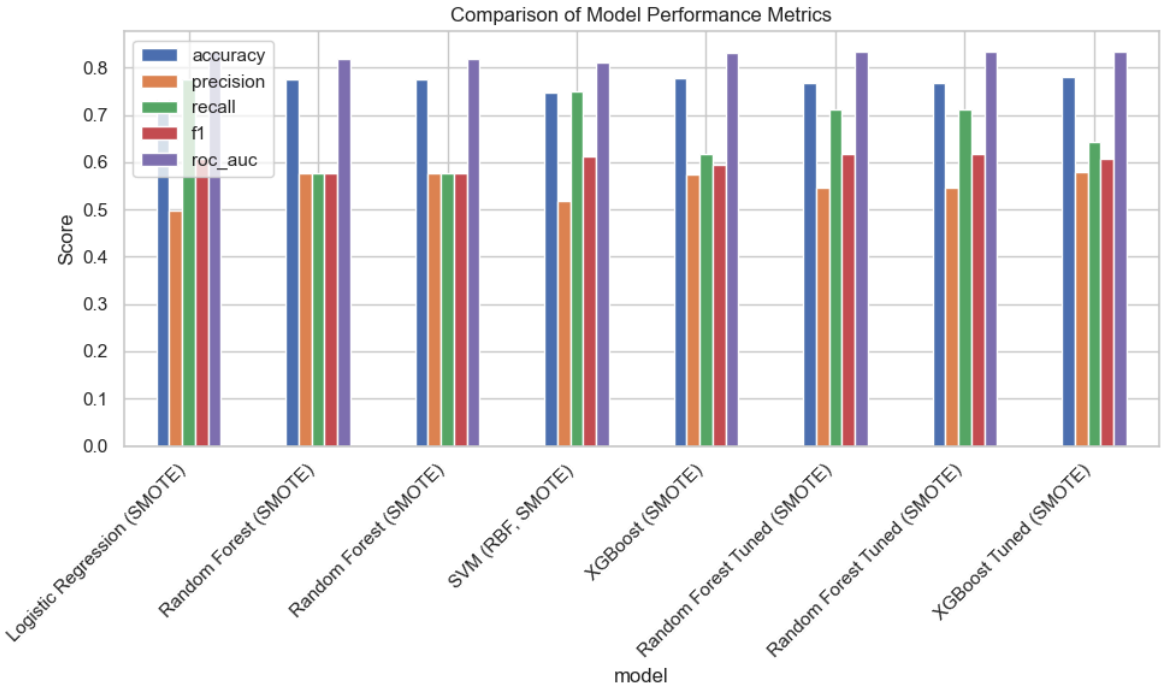
In [ ]:
```
# Bar charts: Precision, Recall, F1-score per model

metrics_to_plot = ['precision', 'recall', 'f1']
colors = ['#4c72b0', '#55a868', '#c44e52']
```

```
plt.figure(figsize=(8, 5))
bar_width = 0.2
indices = np.arange(len(results_df['model']))

for i, (metric, color) in enumerate(zip(metrics_to_plot, colors)):
    plt.bar(indices + i * bar_width, results_df[metric],
            width=bar_width, color=color, label=metric.capitalize())

plt.xticks(indices + bar_width, results_df['model'], rotation=45, ha='right')
plt.ylabel('Score')
plt.title('Model Comparison: Precision, Recall, F1-score')
plt.legend()
plt.tight_layout()
plt.show()
```



In [ ]:

```
metrics_to_plot = ['accuracy', 'precision', 'recall', 'f1', 'roc_auc']

results_df.set_index('model')[metrics_to_plot].plot(kind='bar', figsize=(10, 6))
plt.title('Comparison of Model Performance Metrics')
plt.ylabel('Score')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



In [ ]:
```
results_df
```

Out[ ]:

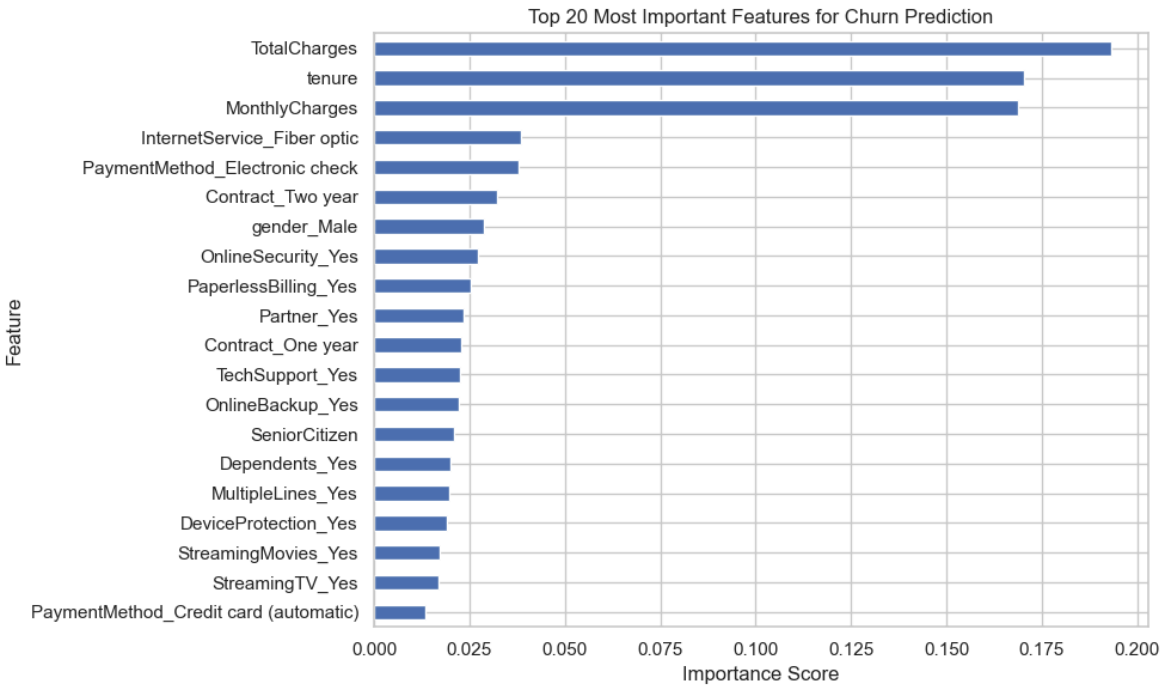
	model	accuracy	precision	recall	f1	roc_auc
0	Logistic Regression (SMOTE)	0.732765	0.498282	0.775401	0.606695	0.833206
1	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
2	Random Forest (SMOTE)	0.775409	0.577540	0.577540	0.577540	0.817588
3	SVM (RBF, SMOTE)	0.747690	0.517560	0.748663	0.612022	0.810739
4	XGBoost (SMOTE)	0.776830	0.574627	0.617647	0.595361	0.831046

In [ ]:
```
# Feature importance using Random Forest on one-hot encoded data

# For feature importance we need explicit one-hot encoded feature names
X_encoded = pd.get_dummies(X, drop_first=True)

X_train_enc, X_test_enc, y_train_enc, y_test_enc = train_test_split(
    X_encoded, y, test_size=0.2, random_state=42, stratify=y
)

rf_for_imp = RandomForestClassifier(n_estimators=300, random_state=42)
rf_for_imp.fit(X_train_enc, y_train_enc)

importances = rf_for_imp.feature_importances_
feature_importance = pd.Series(importances, index=X_encoded.columns).sort_values

print("Top 20 most important features for churn prediction:")
feature_importance.head(20)
```

```
Top 20 most important features for churn prediction:
```

Out[ ]:
```
TotalCharges                            0.193238
tenure                                  0.170365
MonthlyCharges                          0.168902
InternetService_Fiber optic             0.038595
PaymentMethod_Electronic check          0.037837
Contract_Two year                       0.032238
gender_Male                             0.028753
OnlineSecurity_Yes                      0.027205
PaperlessBilling_Yes                    0.025457
Partner_Yes                             0.023462
Contract_One year                       0.022874
TechSupport_Yes                         0.022659
OnlineBackup_Yes                        0.022333
SeniorCitizen                           0.021001
Dependents_Yes                          0.020104
MultipleLines_Yes                       0.019827
DeviceProtection_Yes                    0.018971
StreamingMovies_Yes                     0.017321
StreamingTV_Yes                         0.017025
PaymentMethod_Credit card (automatic)   0.013425
dtype: float64
```

In [ ]:
```
top20 = feature_importance.head(20)

plt.figure(figsize=(10, 6))
top20.sort_values().plot(kind='barh')
plt.xlabel('Importance Score')
plt.ylabel('Feature')
plt.title('Top 20 Most Important Features for Churn Prediction')
plt.tight_layout()
plt.show()
```



In [ ]:
```
import pandas as pd
import shap
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
```

In [ ]:
```
X = df.drop('Churn', axis=1)
y = df['Churn']
```

In [ ]:
```
X_encoded = pd.get_dummies(X, drop_first=True)
```

In [ ]:
```
X_train_enc, X_test_enc, y_train_enc, y_test_enc = train_test_split(
    X_encoded, y, test_size=0.2, random_state=42, stratify=y
)
```

In [ ]:
```
rf_for_imp = RandomForestClassifier(n_estimators=300, random_state=42)
rf_for_imp.fit(X_train_enc, y_train_enc)
```

Out[ ]:
```
▾ RandomForestClassifier
▸ Parameters
```

In [ ]:
```
print(results_df.iloc[0]["Model"])
```

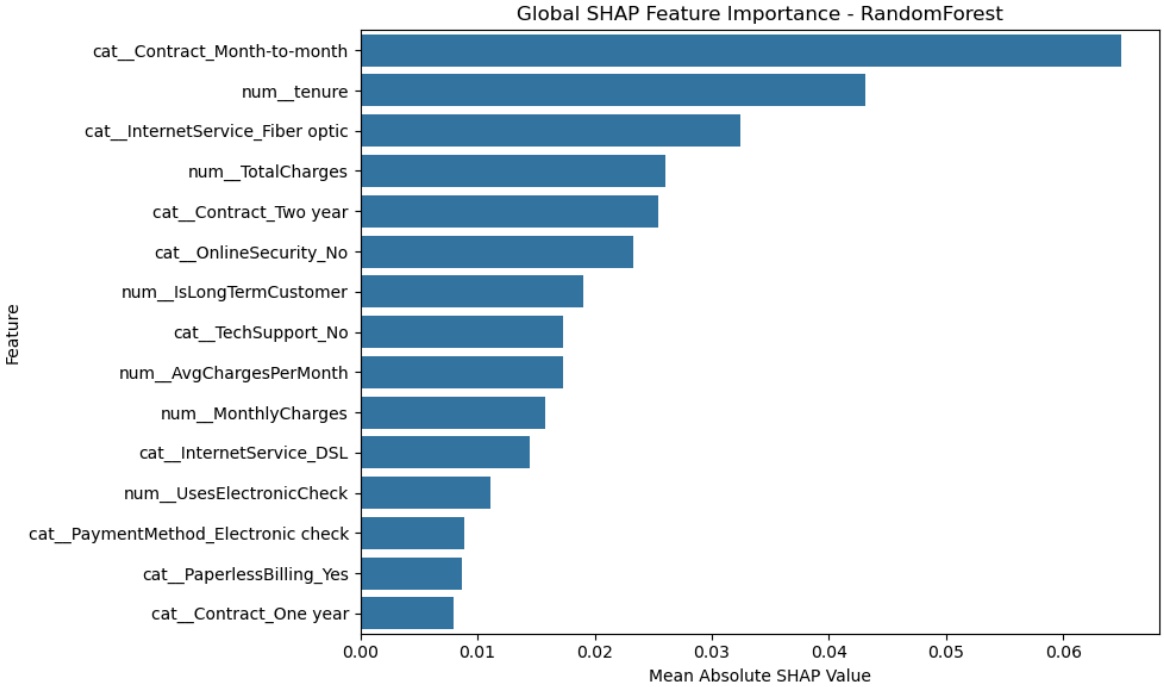
```
RandomForest
```

In [ ]:
```
import shap
import pandas as pd

best_model_name = results_df.iloc[0]["Model"]
best_model = best_estimators[best_model_name]

print("Best model:", best_model_name)

preprocessor_fitted = best_model.named_steps["preprocess"]
clf = best_model.named_steps["model"]

X_test_enc = preprocessor_fitted.transform(X_test)
feature_names = preprocessor_fitted.get_feature_names_out()
X_test_enc_df = pd.DataFrame(X_test_enc, columns=feature_names)

explainer = shap.Explainer(clf, X_test_enc_df)
shap_exp = explainer(X_test_enc_df)

print(type(shap_exp))
print(X_test_enc_df.shape)
print(shap_exp.values.shape)
```

```
Best model: RandomForest
100%|===================| 2809/2818 [01:32<00:00]
<class 'shap._explanation.Explanation'>
(1409, 50)
(1409, 50, 2)
```

In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# get SHAP values matrix
shap_matrix = shap_exp.values

# if SHAP returns 3D output, select the positive-class / main output safely
```

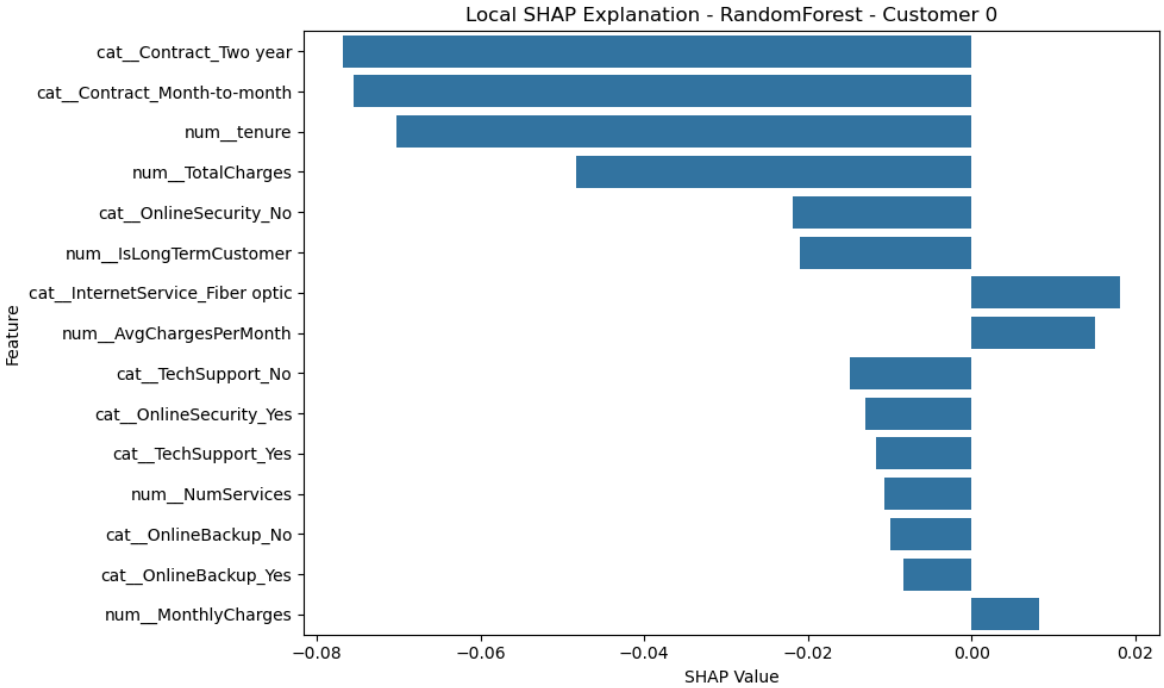
```
if len(shap_matrix.shape) == 3:
    shap_matrix = shap_matrix[:, :, 1]

# make sure feature count matches
n_features = min(shap_matrix.shape[1], len(feature_names), X_test_enc_df.shape[1

shap_importance_df = pd.DataFrame({
    "Feature": feature_names[:n_features],
    "MeanAbsSHAP": np.abs(shap_matrix[:, :n_features]).mean(axis=0)
}).sort_values(by="MeanAbsSHAP", ascending=False)

top_shap = shap_importance_df.head(15)

plt.figure(figsize=(10, 6))
sns.barplot(data=top_shap, x="MeanAbsSHAP", y="Feature")
plt.title(f"Global SHAP Feature Importance - {best_model_name}")
plt.xlabel("Mean Absolute SHAP Value")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```



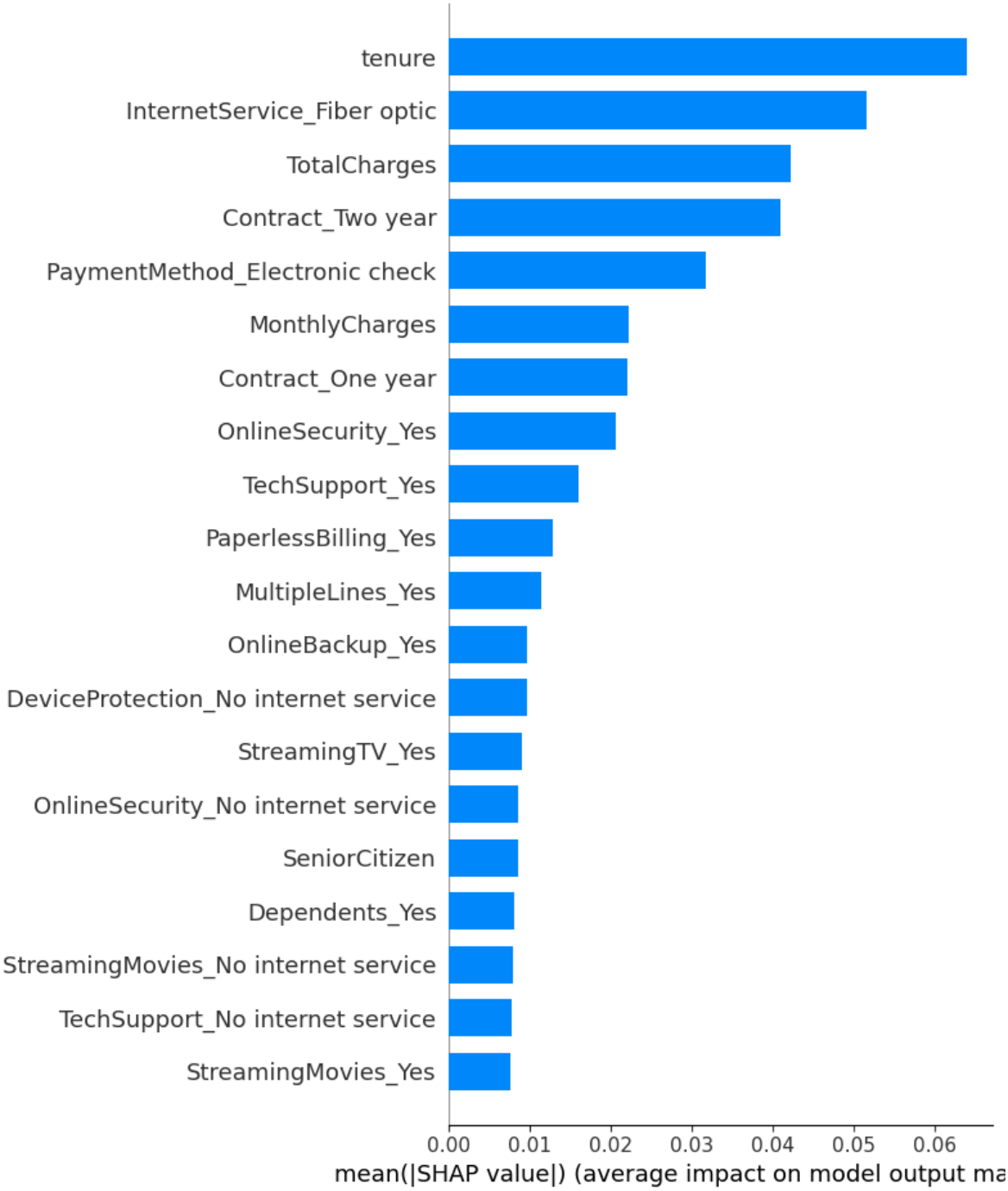
In [ ]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sample_index = 0

local_shap = shap_exp.values[sample_index]

# if 2D for one sample with outputs, keep positive class
if len(local_shap.shape) == 2:
    local_shap = local_shap[:, 1]

n_features = min(len(local_shap), len(feature_names))

local_df = pd.DataFrame({
    "Feature": feature_names[:n_features],
```

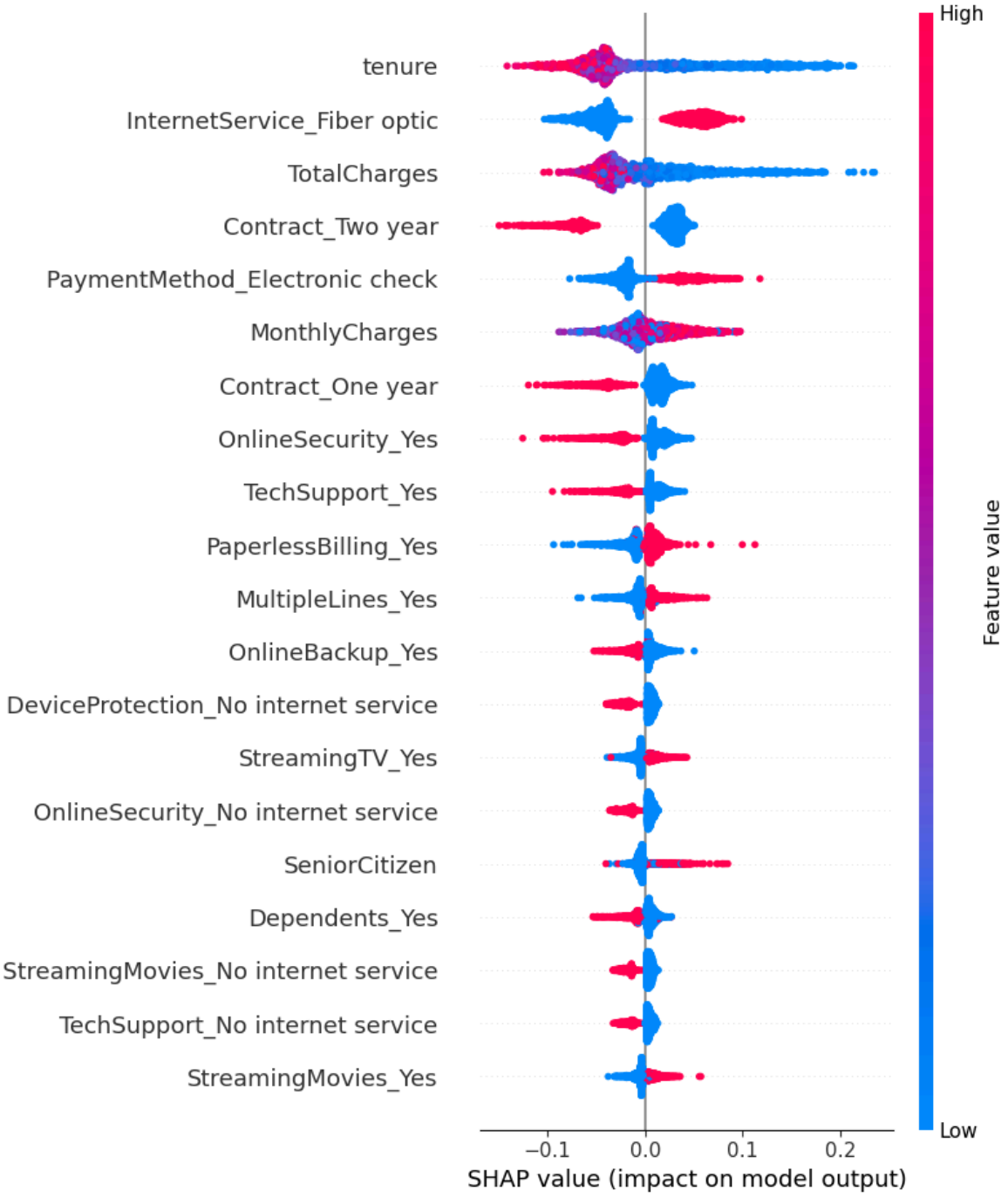
```
    "SHAPValue": local_shap[:n_features]
})

local_df["AbsSHAP"] = np.abs(local_df["SHAPValue"])
local_top = local_df.sort_values(by="AbsSHAP", ascending=False).head(15)

plt.figure(figsize=(10, 6))
sns.barplot(data=local_top, x="SHAPValue", y="Feature")
plt.title(f"Local SHAP Explanation - {best_model_name} - Customer {sample_index}
plt.xlabel("SHAP Value")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```



In [ ]:
```
shap.summary_plot(shap_values[:, :, 1], X_test_enc, plot_type="bar")
```

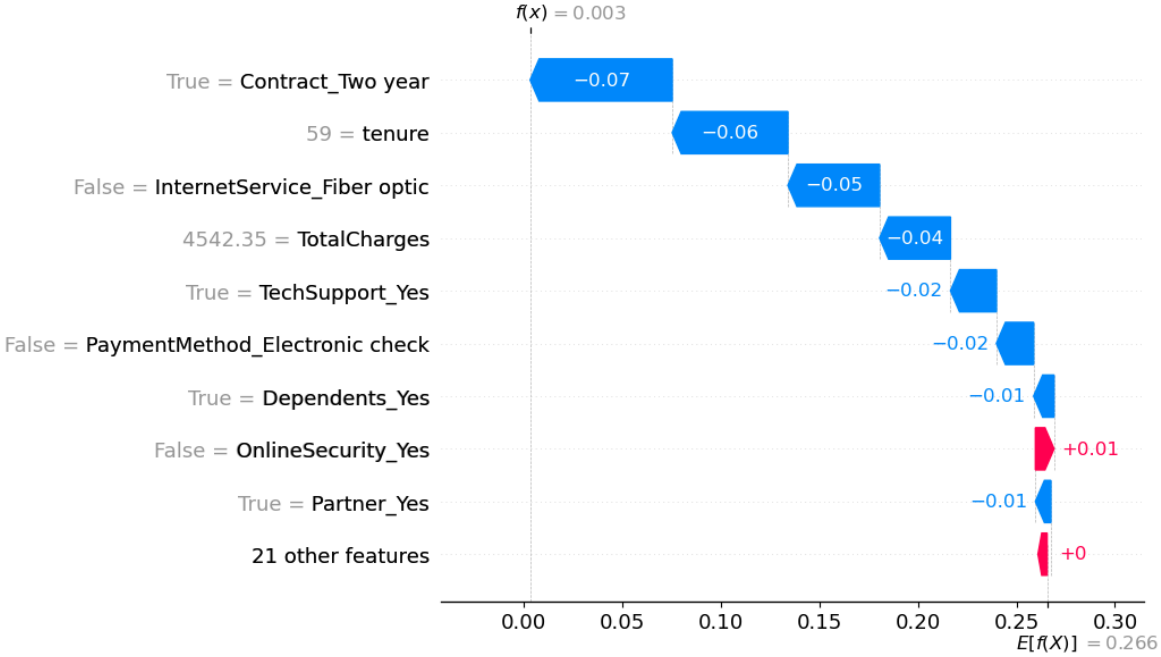


In [ ]:
```
shap.summary_plot(shap_values[:, :, 1], X_test_enc)
```



In [ ]:
```
sample_idx = 0

if isinstance(shap_values, list):
    local_sv = shap_values[1][sample_idx]
    base_val = explainer.expected_value[1]
elif len(shap_values.shape) == 3:
    local_sv = shap_values[sample_idx, :, 1]
    base_val = explainer.expected_value[1] if isinstance(explainer.expected_valu
else:
    local_sv = shap_values[sample_idx]
    base_val = explainer.expected_value

shap.plots.waterfall(
    shap.Explanation(
        values=local_sv,
        base_values=base_val,
        data=X_test_enc.iloc[sample_idx],
        feature_names=X_test_enc.columns
```

```
    )
)
```


f(x) = 0.003
True = Contract_Two year
59 = tenure
False = InternetService_Fiber optic
4542.35 = TotalCharges
True = TechSupport_Yes
False = PaymentMethod_Electronic check
True = Dependents_Yes
False = OnlineSecurity_Yes
True = Partner_Yes
21 other features
−0.07
−0.06
−0.05
−0.04
−0.02
−0.02
−0.01
+0.01
−0.01
+0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
E[f(X)] = 0.266